Band
1

Induction Heating System Series i-class compact

EN

Operating Manual

Stand: 800 0031.02, 16.10.2025

Note!

This operating manual is required for the safe operation of the induction generator. As such, the manual must be kept in close proximity to the device.

COBES GmbH shall not be liable for any mistakes in this documentation. Liability for direct and indirect damage arising from the delivery or use of this documentation shall be excluded as far as is legally permissible.

Every attempt has been made to describe the product as accurately as possible within this documentation. However, no warranty about specific properties or the outcomes of operating the device is implied. We always welcome notifications of any errors as well suggestions and comments.

Unless otherwise stated, the technical specification of the product at the time of its combined delivery with the documentation by COBES GmbH shall apply. We reserve the right to make technical changes without prior notification. Previous documentation shall cease to be valid in such a case.

Please call us if you have any questions or problems with the installation and start-up. We'll be only too happy to help you.

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INDUCTION HEATING SYSTEM

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Introduction

This operating manual describes a medium-frequency generator for the inductive heating of electrically conductive workpieces. For this purpose, the induction generator generates an alternating magnetic field in an induction coil at a frequency of MF 20 - 100 kHz.
The induction generator consists of the generator itself and the induction coil, which is connected to the generator.
These components must be supplied with cooling water from a cooling unit.

To prevent incorrect operation and damage to both the operator and the device, the operating instructions and safety instructions must be read and observed before commissioning the induction generator to ensure safe use.
All persons entrusted with the use of the induction generator must also be familiar with the operating instructions and observe them before using the device for the first time.
Keep the operating instructions in a safe place so that the information they contain is available to you at all times.

No excerpts from these operating instructions may be reproduced, stored in a data processing system, or transmitted without the prior written consent of COBES GmbH.

COBES GmbH reserves the right to make technical and visual changes at any time without prior notice in the course of further development. All dimensions, information, and specifications in this operating manual are therefore subject to change without notice. Legal claims based on the operating manual cannot therefore be asserted.

INDUCTION HEATING SYSTEM

1. Scope of application

1.1 General information

This operating manual applies to the i-class compact (ICC) series induction generator and the associated connecting cables.

1.2 Device designation and type

Device designation	Device type	Serial No.
Induction heating generator Serie i-class compact	1 - 5kW MF 20 - 100 kHz	20JJ-MM-XXXX

1.3 EU Declaration of Conformity

We, COBES GmbH, Rudolf-Hell-Str. 8A, D-77955 Ettenheim, hereby declare that the above-mentioned device complies with the relevant basic safety and health requirements of the EU Directive due to its design and construction as well as the version we have placed on the market. This declaration loses its validity in the event of any changes not agreed with us. The EMC-compliant design refers only to the designated device. If the device is expanded to form a system, the operator is responsible for compliance with EMC requirements and personal safety.

1.4 Directives and standards

EU Directives:

2014 / 35 / EU	Directive of the European Parliament and of the Council of February 25, 2014, on the harmonization of the laws of Member States relating to the making available on the market of electrical equipment designed for use within certain voltage limits (Low Voltage Directive)
2014 / 30 / EU	Directive of the European Parliament and of the Council of February 26, 2014, on the harmonization of the laws of Member States relating to electromagnetic compatibility (EMC Directive)

Standards taken into consideration:

DIN EN 60519-1:2020-12	Safety in electrical heating installations and installations for electromagnetic processing Part 1: General requirements
DIN EN 60519-3:2018-03	Part 3: Special requirements for inductive and conductive heating and processing equipment
DIN EN 61010-1:2020-03	Safety requirements for electrical measuring, control, and laboratory equipment Part 1: General requirements
DIN EN 61000-6-2:2020-05	Electromagnetic compatibility (EMC) - Part 6-2: Generic standards - Immunity for industrial environments
DIN EN 61000-6-4:2021-10	Electromagnetic compatibility (EMC) - Part 6-4: Generic standards - Emission standard for industrial environments

1.5 Warranty

COBES devices are manufactured and undergo final inspections using pretested material. If, however, grounds for a complaint exist, the material and manufacturing defects are covered by a warranty usually extending for a period of 12 months from the date of purchase. Wearing parts, particularly hose cables, are not covered by the warranty.

Exclusion of warranty

Damage caused by improper use or tampering with the device is excluded from this warranty.

2. Instructions for Safe operation

2.1 Warning notices

Grading of the warning notices

Warning notices are prefixed by the following signal words denoting the type of danger:

➔ Caution warns of material damage.
➔ Warning warns of personal injury.
➔ Danger warns of fatal injuries.

Structure of the warning signs

Signal word

WHQuelle
➲ WHMaßnahme

2.2 Additional symbols

Explanation of symbols

Symbol	Text	Explanation
	Note!	Important information in the operating manual on optimal use of the device.
	Attention!	This symbol warns of dangers to the product. Failure to heed this warning may result in material damage and, in extreme cases, in the complete destruction of the product and adjoining parts of the plant.
	Electric shock!	This symbol warns of dangers from electric voltage. Failure to heed this warning may result in an electric shock with all the known injuries this can cause, including fatal injuries.
	Electromagnetic fields!	This symbol warns of dangers from electromagnetic fields.
	High frequency!	These signs warn of dangers posed by high-frequency radiation. People with pacemakers or implants, as well as pregnant women, must not remain in areas marked in this way, as high-frequency radiation can interfere with the functioning of pacemakers and cause implants to heat up.

Handling instructions

Structure of the instructions:
➲ Instructions on handling.
Indication of an outcome, if necessary.

Lists

Structure of unnumbered lists:

- List level 1
  → List level 2

Structure of numbered lists:

1. List level 1
2. List level 1
   2.1 List level 2
   2.2 List level 2

2.3 Operator obligations

In addition to complying with the statutory provisions that the manufacturer must comply with, the operator must also comply with and implement the legal obligations in force at the workplace e.g. the duty to provide instructions, the legislation concerning health and safety at work as well as other regulations and laws in force. The operator must draw up written instructions for working on and
with the plant based on the operating manual and the work to be carried out. These instructions must be available in a comprehensible form in the employees' own language and must be made known to them.

2.4 Maintenance book

We recommend that a maintenance book be kept for recording inspections as well as all essential work (e.g. repairs, modifications). Keep this operating manual in a safe place so that the safety instructions and important instructions for use it contains are always available for reference.

2.5 Safety instructions

The safety instructions relating to the protection of persons, handling of the device, and the workpiece are largely dependent on the behavior of the persons working with the device. Before commissioning, read the operating instructions carefully and observe the information provided in order to avoid errors and any resulting damage, in particular damage to health. In addition to the information in this operating manual, the applicable national laws, regulations, and guidelines must be observed for installation and operation.
Work on the electronic equipment of the device may only be carried out by trained service personnel and in a safe condition (de-energized). Only approved original spare parts and accessories may be used. The use of other parts involves unknown risks and must be avoided in all cases.
The functionality and safety of the device is only guaranteed if the necessary tests, maintenance, and repair work are carried out by COBES or by trained personnel.
COBES is not liable for any damage to the device resulting from improper repairs not carried out by COBES, or if original spare parts/accessories are not used when replacing parts.

2.6 Dangers in the operating room

The emission of electromagnetic fields depends on the application. In principle, the operator of the system is responsible for the safety of persons in electromagnetic fields (EMF). We are happy to assist with on-site measurements in accordance with the applicable regulations of the relevant employers' liability insurance association.

Danger!

Alternating electromagnetic fields can have an undesirable, harmful effect on active and passive body aids as well as future life (pacemakers, implants, pregnancy, etc.).
Persons with active and passive body aids and pregnant women must not be in the area affected by the device.
➲ Attach the supplied adhesive labels "Prohibition for pacemaker wearers", etc. to all devices that are operated with induction

Danger!

Strong electromagnetic fields occur in the area of the inductor coil. The user is responsible for compliance with the permissible limit values at the installation site. The guidelines applicable at the installation site apply.

2.7 Danger due to electrical voltage

Danger!

The induction generator converts low-frequency electrical mains energy into medium-frequency electrical energy. The resulting voltages are life-threatening!
➲ Never open the generator while it is connected to the mains supply.

Danger!

Dangerous voltages!
Dangerous voltages are present at the inductor when the generator is switched on.
➲ Fit suitable covers and safety devices to protect against contact!

Attach a suitable warning sign with the five symbols in the area of the inductor coil:

Illustration 1: Warning signs for induction systems

2.8 Danger points during maintenance and repair

The induction generator must be connected by the system manufacturer to a 3-phase mains supply with a protective earth conductor in accordance with the applicable regulations.
There are no wearing parts or parts requiring maintenance in the generator itself. The generator must therefore be sent to the manufacturer for repairs.

Danger!

Dangerous voltages!
As soon as the generator is connected to the supply voltage, dangerous voltages may be present in the generator.
➲ Before opening the generator, remove all plugged-in cables.
➲ The generator may only be opened by the manufacturer!

2.9 Safety devices

Note!

As the operator, it is your responsibility to ensure that the HF generator operates safely. You are responsible for familiarizing yourself with and applying the statutory provisions that must be complied with.

Personal protection
In induction applications, the induction coil (inductor) must be protected from contact by the operator, as a dangerous voltage is present there when the generator is in operation.

Safety circuit
To switch off the output power of the generator, the system manufacturer must ensure that the mains connection to the generator is interrupted in the event of a fault. The need to integrate additional switching devices depends on the desired performance level. A suggestion for the design can be found in chapter 6.3 “Connecting”.

System protection
To protect the generator itself and connected devices, the generator switches off the output power in the event of:

- Overtemperature
- cooling water flow is missing/low
- Safety circuit error (interlock open)
- Communication error in serial operation

Please note that this only switches off the internal electronics and the MF voltage. This is not a safety function that affects protection against accidental contact. The generator must be disconnected from the mains for personal protection.

2.10 Emergency measures

In an emergency, take the following measures immediately in the order described:

- 1. switch off the external mains disconnecting device for the generator. (PL at least D)
- 2. switch off the cooling water supply for the generator.
- 3. extinguish fires with suitable extinguishing agents.

Danger!

➲ Do not extinguish fires with water, due to the mains voltages present, but use suitable extinguishing agents.

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3. Intended use

The induction generator generates process energy that is introduced into the workpiece via a current-carrying coil, the inductor. Excitation of the inductor at medium frequency (MF) generates an alternating magnetic field. If a conductive workpiece is in this field, it is heated with a high energy density. Depending on the frequency of the alternating field, the workpiece only heats up on the surface. All electrically conductive and semi-conductive materials can therefore be heated by alternating electromagnetic fields. Up to the Curie point (loss of magnetic properties), which is approx. 730°C for steel, magnetic materials can be heated particularly well by induction. For temperatures above the Curie point, significantly more generator power must be installed. This also applies to highly conductive materials such as copper, aluminum, etc. The higher the operating frequency of the generator, the less deeply the field lines penetrate the material. With longer exposure times, however, heating can also be achieved due to heat conduction in the workpiece. In order to heat a so-called surface layer (e.g. hardening of steel), the workpiece is heated very quickly above the structure transformation temperature and then cooled just as quickly with a quenching spray (usually an oil-water mixture emulsion). This results in a narrow-hardened zone (surface layer) on the one hand, and prevents the transformation of the microstructure in the rest of the workpiece on the other, thus maintaining the hardness there.

The most common applications are Bonding, joining, soldering, melting, forging, hardening, pre/post-heating, tempering, sealing and many more.

3.1 Impermissible modes of operation

The induction generator may only be used for its intended purpose. In particular, it may not be used

- for medical purposes,
- for radio engineering purposes,
- in rooms with an explosive atmosphere

3.2 Work rules

Attention!

➲ If particular problems occur that are not covered in sufficient detail in these instructions for use, please contact the supplier for your own safety.

Attention!

➲ Put on the required personal protective equipment, e.g. hand, face and body protection, remove any jewelry beforehand.

Attention!

Only use the system for its intended purpose.
In particular, it is not permitted to use:
➲ for medical technology purposes,
➲ for radio technology purposes,
➲ in rooms with explosive atmospheres

4. Scope of delivery

Part number	Designation/model	Quantity
126 00xx	Induction generator 5kW, 20-100kHz, xx Wdg.	1
126 0075	Induction generator 5kW, 20-100kHz, 28 Wdg.	
126 0081	Induction generator 5kW, 20-100kHz, 18 Wdg.	

Upon delivery of the device, check that the components are in perfect condition. Please contact COBES if you have any complaints.

4.1 Spare parts and accessories

Part number	Designation/model	Quantity
559 0103	O-Ring, d=6x2,0, NBR, 70A	
552 0041	Scheibe, Fächer-, DIN 6798, A5,3, Bronze	
550 0217	Schraube, Zylinder-, Inn-6kt, M5x20, DIN 912, A2	

4.2 Options

Part number	Designation/model	Quantity
309 0003	**Zubehörpaket, ICC, Standard**	**1**
575 0110	Stecker, Tüllengehäuse HAN 3 A, gerade, M20	1
562 0011	Verschraubung, Kabel-, M 20x1,5 EMV, 8-11	1
251 0178	Kabel, 4G2.5, Ø=8.9, grau, Ölflex	3m
575 0066	Buchse, Kontakt 2,5mm² gedr., HAN C, Crimp Female	4
575 0106	Stecker, Einsatz, 4-pol., Han Q3/0-M-Crimp	1
571 0018	Stecker, Netz, CeKon, 5-polig, 16A	1
564 0093	Leiterplatte, Adapter	1
563 0135	Kabel, Patchkabel, M8, RJ45, 2m	1
563 0147	Kabel, Anschluss-, D-SUB 15 pol., St.- Bu., 2m	1
309 0004	**Anschraub-Set, Induktor, an ICC**	**1**
559 0103	O-Ring, d=6x2,0, NBR, 70A	3
552 0041	Scheibe, Fächer-, DIN 6798, A5,3, Bronze	4
550 0217	Schraube, Zylinder-, Inn-6kt, M5x20, DIN 912, A2	4
309 0005	**Zubehörpaket, ICC, Wasser**	**1**
661 0018	Sensor, Durchfluss-, Wasser, 0,9-15l/min	1
651 0112	Verschraubung, Überwurf-, 3/4" IG auf 1/2" AG	2
651 0088	Muffe, Innengewinde, rund, G1/2"	2
651 0113	Verschraubung, Einschraub-, G1/2", 8mm, gerade	2
661 0019	Strömungswächter, Anschlussk. mit Buchse M12, 2m	1
657 0025	Schlauch,TPE-U(PU), 8x1,25 transparent, außenkal.	6m

4.3 Necessary requirements on the customer side

The following requirements must be met by the customer before the generator can be put into operation.

Cooling water supply:
A suitable cooling water supply must be ensured. The following values must be observed:

- Permissible cooling water temperature in the flow: 20 to 40 °C
- Permissible flow rate: 1 to 8 l/min
- Permissible maximum pressure in the flow: 8 bar
- Differential pressure at the generator: approx. 3.5 bar (depending on the inductor)
- Outer diameter of hose: 8mm

Power supply:
A 400V, 16A three-phase connection (AC 3~In) is required to power the generator.
In addition, a 24V ± 10 %, max. 3.5A power connection is required.

Control:
Digital communication see chapter Quick overview of process data (UDP control)

INDUCTION HEATING SYSTEM

5. Description of the device

5.1 Components and their function

The induction system essentially comprises the following components:

- Generator with connection for the inductor
- Inductor, for heating the workpiece

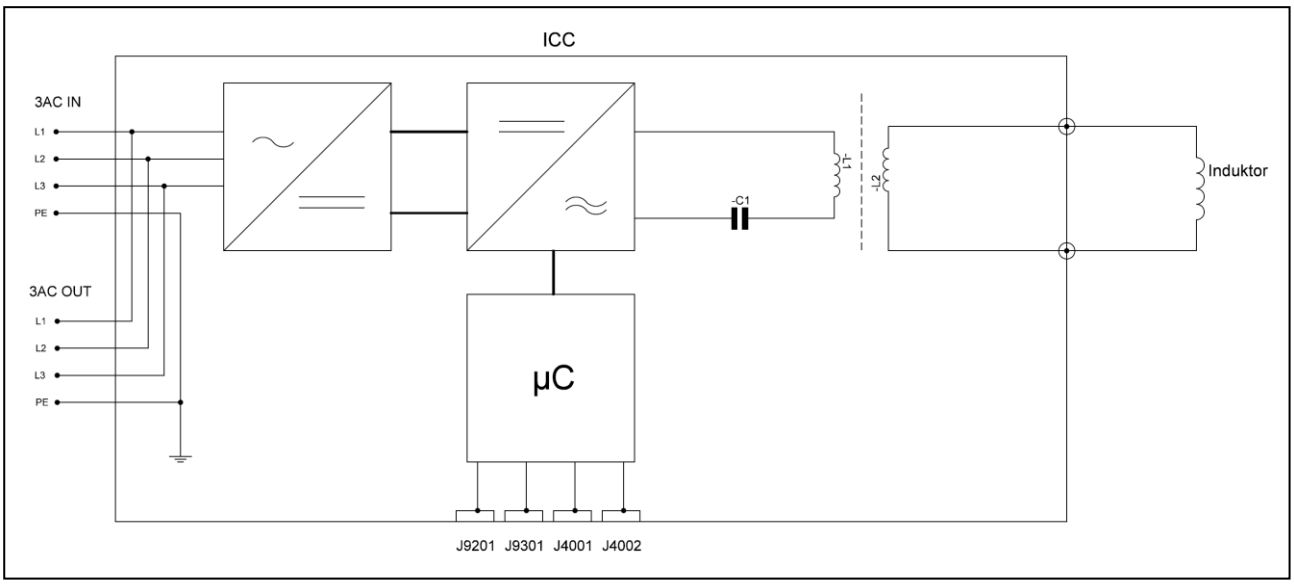


Illustration 2: Schematic diagram of the system ready for operation

The mains input voltage is fed to the rectifier via the 3AC IN mains connection. The inverter uses this DC voltage to generate the AC voltage in the range of 20-100 kHz, which is fed to the inductor via a transformer. The output power can be continuously adjusted between approx. 5 and 100%. The power control in the device is achieved via a variable frequency.
For maximum power output, the system consisting of the generator-inductor-workpiece must be optimally adjusted. If the adjustment is poor, the power output can be drastically reduced. COBES will be happy to assist you in adapting to new workpieces.

5.2 Range of applications

Typical areas of application

The induction generator is designed as a stationary energy source for the inductive heating of electrically conductive materials at medium frequency. Typical areas of application are

- Bonding
- Joining
- Soldering
- Melting
- Forging
- Hardening
- Pre/post-heating
- Tempering
- Sealing

5.3 Technical data

Detailed information can be found in the enclosed documents technical specification and Cooling water specification

5.4 Options

COBES can support you in the procurement or dimensioning of a suitable cooling unit for the cooling water required in the generator. Please contact our staff for more information.

6. Transport, set up, installation and initial start-up

6.1 Transport and storage

Transport

The transportation of the generator from the delivery vehicle to the final installation site must be prepared and carried out by the customer. After receipt of the generator:

- Immediately notify the carrier, the insurance company and COBES in writing of any transportation damage due to recourse claims.

The induction generator is delivered in a cardboard box. It is protected with foam inserts and filling material during transportation.

Storage

If you do not install the components immediately after delivery, store it in the original packaging under the following ambient conditions:

Temperature

- +10 °C to +45 °C Operation
- –20 °C to +55 °C Storage
- –20 °C to +70 °C Transport

Relative humidity

- 10 % – 90 % Operation[1]
- 30 % – 95 % Storage
- 95 % Transport

Air pressure2[2]

- 800 hPa bis 1 013 hPa Operation
- 800 hPa bis 1 013 hPa Storage
- 660 hPa bis 1 013 hPa Transport

1) No thawing and freezing.
2) Max. 2,000 m above sea level.

6.2 Unpacking

Unpack the components of the HF generator carefully. The components are delivered with all of the connections required for the initial start-up already made.

Packaging material

Dispose of the packaging materials in accordance with the relevant waste disposal regulations unless you intend to save them to transport the device at a later date. The following materials are used as packaging depending on the type of transport:

- Plastic foil (PE) to protect the device
- Cartons
- Foam inserts and foam padding

All packaging materials must be disposed of in accordance with the relevant waste disposal regulations applying to the area where the device is delivered.

6.3 Connection

Danger!

Dangerous voltage!
Touching the supply voltage can result in fatal injuries! Therefore do not make any connections unless the components have been disconnected from the supply voltage.

Note!

According to EN ISO 13849-1:2015 (transition period until May 15, 2027), safety for man and machine is divided into risk categories or performance levels. This device achieves performance level PLd with an upstream isolating device.

The electrical connection may only be carried out by qualified and trained electricians. These specialists must be informed about the applicable safety regulations
at the installation site for setting up electrical installations and apply them. Improper installation can result in danger to life due to electrical voltages and leakage currents!

If you include an emergency stop switch, you must install it in such a way that it triggers the external mains disconnection device in accordance with PLd when actuated.
The disconnection of the cooling water supply depends on the respective application.

6.3.1 Procedure

Always connect the cooling water supply first and check it for leaks before continuing with the electrical installation.

Proceed in the following order:

- Connect the inductor
- Connect the cooling water supply
- Connecting the control circuits
- Connecting the power supply

6.3.2 Connecting the inductor

Place the supplied sealing rings (see illustration) between the inductor and the inductor connection jaw and screw the inductor to the inductor connection jaws. Only use the screws supplied for this purpose. Never use conventional steel screws! The recommended tightening torque is 2 Nm.

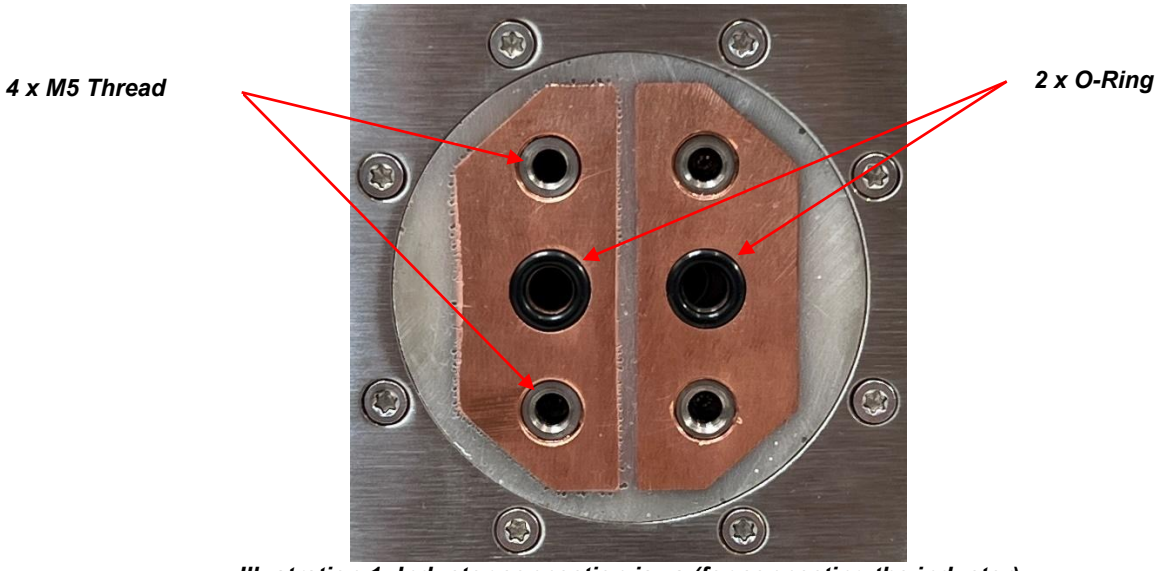


Illustration 1: Inductor connection jaws (for connecting the inductor)

Attention!

- ➲ Make sure that the O-rings supplied are correctly inserted in the inductor connection jaws.
- ➲ Never use steel screws to mount the inductor!
- ➲ The operator must protect the inductor from contact!

6.3.3 Connecting the cooling water supply (Note the cooling water supply!)

Connect the cooling water hoses to the generator as shown in the following illustration. Make sure that you do not mix up the supply and return lines. Also observe the cooling water plan and the cooling water quality requirements.

- Connect the cooling water supply to the “cooling water supply” cooling water connection
- Connect the return line of the cooling water to the “Cooling water return” connection
- Connect the cooling water return line to the flow sensor
- Connect the flow sensor to the cooling water supply

Permissible cooling water temperature in the flow: 20 to 40 °C
Permissible volume flow: 1 to 8 l/min
Permissible maximum pressure in the flow: 8 bar

Illustration 2: Cooling water connections on the generator

Illustration 3: Cooling water connection on the flow sensor

The connections are designed for externally calibrated TPE-U(PU) 8x1.25mm and only need to be plugged in.

After the cooling water connections have been made, check that the cooling system is tight:

- Before switching on the cooling water supply, check again that all water connections are tight.
- Switch on the cooling water supply slowly until the generator is completely filled with water and the system is free of air.
- Immediately after switching on, check all connections and water screw connections for leaks.
- Switch the cooling water supply off again.

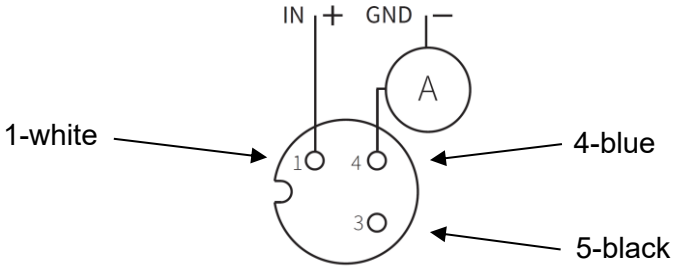
The flow connections of the generator should be fitted with a water filter. Only operate the generator with water filters, otherwise dirt particles may enter the cooling system and cause damage!

Note!

➲ The operator must ensure that the appliance is water-cooled.
➲ Only operate the generator via a dirt particle filter.
➲ The flow monitor must always be installed in the return flow.

6.3.4 Pin assignment of flow sensor connection cable

Cable: IFM - EVC004



Current output

6.3.5 Connecting control circuits

The user interfaces are located on the rear of the generator.

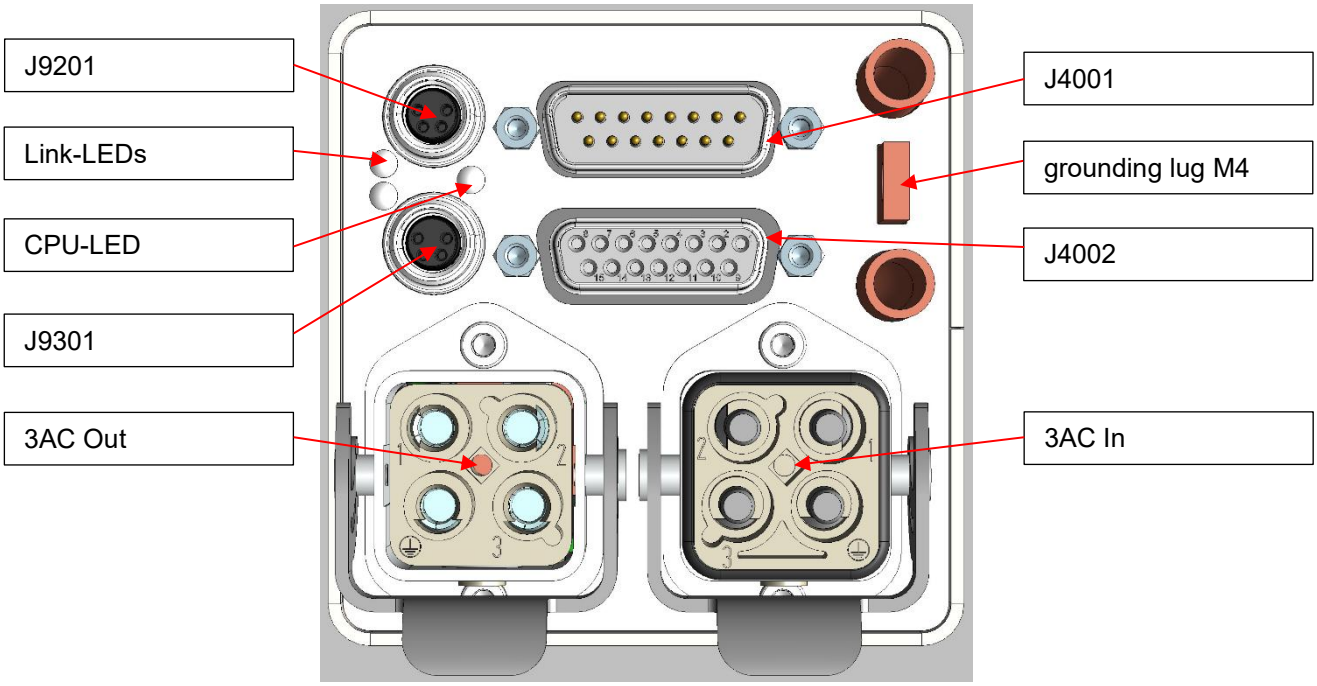


Illustration 6: Connections on the generator

Connecting system plug connectors (supply, control, sensor connections)

Plug connector: D-Sub plug connector, 15-pin
Conec 15-006503 (socket), Conec 15-006453(plug))

Common, looped-through signals on J4001 and J4002:

Pin	Designation	Function
1, 2, 9	24V	Power supply of the control section. 24V ± 10 % Max. Current consumption: 3.5A
3, 10, 11	0V	
4, 12	ILK1, ILK2	Interlock signals (pin 4 → 24V external)

Connector J4001 is used as an input for the following signals:

Pin	Designation	Function
5	FLOW_P	Power supply and sensor input of the flow sensor, FLOW_P against FLOW_N supplies the voltage, the current flowing in FLOW_N represents the measurement signal
13	FLOW_N	
6	PYRO_P	Pyrometer connection, current input (4...20mA interface, internal 100Ω measuring resistor)
14	PYRO_N	Pyrometer connection, signal ground
7	L+_CON	24 volt supply voltage for pyrometer (related to PYRO_N)
8	TC_N_CON	Thermocouple, - connection
15	TC_P_CON	Thermocouple, + connection

Analog input thermocouple
A type K thermocouple can be connected to contacts 8 and 15 in connector J4001 for temperature detection. The polarity of the thermocouple must be taken into account!

Analog input pyrometer
A pyrometer with a 4...20 mA interface can be connected to contacts 6 and 14 in connector J4001 for temperature measurement. 24 volts can be tapped at pin 7 to supply the pyrometer with power.

Connecting the network interface

Plug connector: M8 plug connector, 4-pin, female, A-coded
Phoenix Contact 1412228

Both connectors are interconnected by a 2-port switch to enable network traffic from one port to the other and thus a line topology.
Both ports support 10/100 Mbit/s and HP Auto-MDIX.

Pin assignment on connectors J9201 and J9301:

Pin	Designation	Function
1	TD+	Transmit-Data Differential pair
4	TD-	
2	RD+	Receive-Data Differential pair
3	RD-	

The power cables must be laid in accordance with VDE 0298, Part 4, installation type E.

6.3.6 Connecting the power supply

Danger!

Dangerous voltage!
➲ Before connecting the mains cable, make sure that the disconnecting device in the external installation is open and secured against being switched on again.

Three-phase input (AC 3~ In)

Connectors: Harting Han 3A plug flanges (item no. 09200030301)
AC input with pin contact plug insert (item no. 09120033151)
AC output with socket contact plug insert (item no. 09120033051)

Coding: None

Pin	Designation	Function
1	L1	Three-phase current input 400V
2	L2	
3	L3	
PE	PE	connection Protective earth

Note!

Permissible earthing types: • Neutral conductor to earth potential

The following earthing types are not permitted: • Outer conductor to earth potential
• Insulated (not earthed)

Note!

The mains input is not internally protected against overload.
An external fuse is therefore required to ensure line protection.

Once all electrical connections have been made, check that all plugs and cables are secure.

Equipotential bonding on the generator

There is an earthing lug with a hole for an M4 screw between the cooling water inlets and outlets. An equipotential bonding conductor must be connected here in accordance with DIN VDE 0100, Part 540.

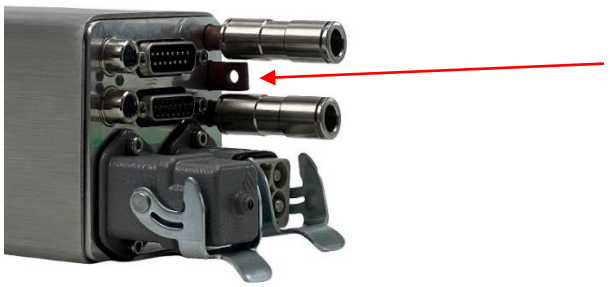


Illustration 4: Equipotential bonding on the generator

6.3.7 Interlock contacts

The following shows two ways in which the two interlock contacts can be wired.

Option 1 (Figure 7) is sufficient for commissioning.
Option 2 (Figure 8) offers a higher level of safety as no floating voltages occur.

Note!
COBES GmbH recommends option 2.

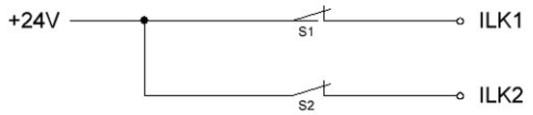


***Illustration* 5: Option 1 Interlock circuit**

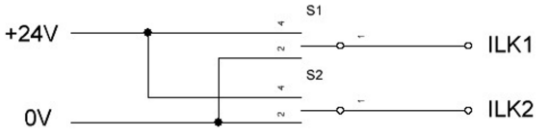


***Illustration* 6: Option 2 Interlock circuit**

Note!
The system manufacturer or operator is responsible for calculating the final performance level, taking into account the connection of the device in the system.

The following measures must be observed when integrating the generator into the system.

The following is designed to ensure a secure hold:
-Blocking of the inverter's gate signals using discrete semiconductor components
(deactivation of the drivers)
-Blocking of the driver signals via the microcontroller control
(software shutdown)
-Disconnection of the driver operating voltage

The following was evaluated for the consideration of dangerous DC voltage on the inductor system:

Due to its design, the output transformer T1 can only transmit high-frequency AC voltages. The transformer is grounded on one side.

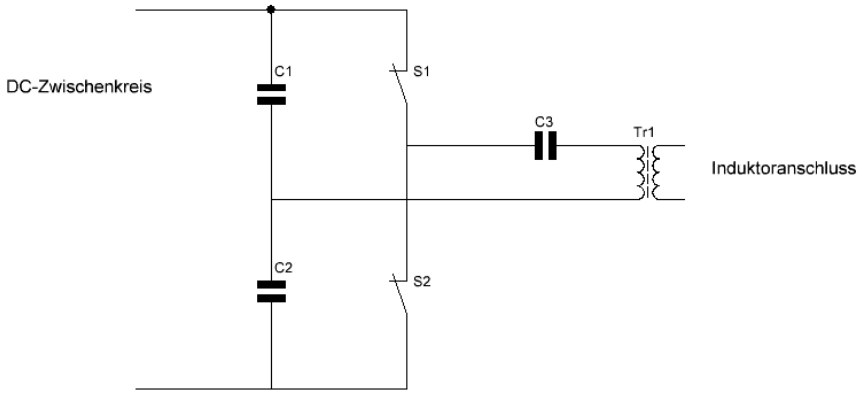


Illustration 7: Schematic diagram of power section

Fault case Inverter bridge (S1 and S2) low impedance:
This leads to a short circuit in the DC link, which triggers the upstream fuse.

Fault case low-resistance semiconductor (S1 or S2) of the inverter bridge:
In this case, there is a DC voltage at the inverter bridge output. The primary capacitor (C3) would then function as a DC capacitor and prevent further currents from flowing through the transformer. Leakage currents in the capacitors can raise the potential of the primary winding to the positive or negative potential of the DC link. This potential difference is isolated by the transformer. If there is an additional insulation fault in the transformer, the currents are discharged via the grounding.

Note!
In case of doubt, COBES GmbH recommends installing a contactor in the supply voltage for safe shutdown.

6.3.9 Connecting additional generators

It is possible to connect up to three generators in series and install them. No additional power connections are required for connecting the generators in series, as these can be looped through from one generator to the next.
Water can also be installed in series, depending on the inductor. For small inductor cross-sections, we recommend connecting the cooling water in parallel. In each configuration, each ICC requires its own water sensor.
A Sub-D cable is plugged into connection J4002 on the first generator and into connection J4001 on the next generator.
The network interface is connected at point J9201 on the first generator and at point J9301 on the second generator.
In order to also supply the second generator with 400V, a connection is established from the first generator (3AC Out) to the second generator (3AC In).

6.3.10 Connecting the secondary capacitor

An additional secondary capacitor is required for certain applications. A special connection clamp is needed to connect this capacitor, which in this case is supplied by COBES. The connection clamp is screwed between the generator and the inductor. The inductor can then be attached as described in section 6.3.2 above.
Here too, make sure that the O-rings are correctly inserted into the generator's inductor connection clamp. The technical drawing for this connection clamp can be found in the appendix.

Attention!

- ➲ Ensure that the supplied O-rings are correctly inserted on the inductor connection jaws.
- ➲ Never use steel screws for assembly!
- ➲ Check the voltage resistance of the capacitors before commissioning!
- ➲ The connection jaw and secondary capacitor must be protected from contact by the operator.

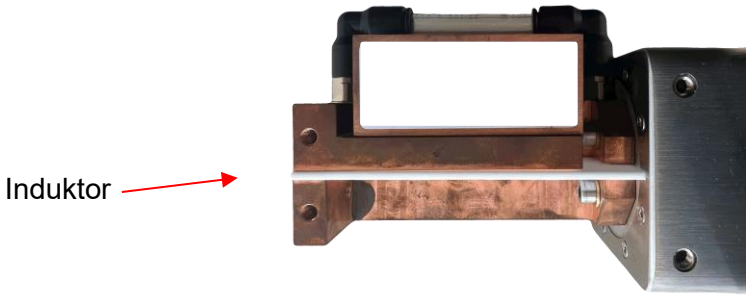


Illustration 8: Connecting the secondary capacitor

6.4 Changing the IP configuration

The IP configuration can only be changed or reset using the “IPConfig” software.
The IPConfig tool allows you to set the IP configuration, enable or disable DHCP, and set the host name of a device.
IPConfig displays the currently set configuration. After changing the settings, the device must be restarted so that the new address configuration is applied.

First, the “IPConfig” program must be installed on the PC from the supplied USB stick. The next step is to connect the ICC generator to the PC via the network cable and connect a 24V power supply to the generator.
Open the IPConfig tool on the computer. Select the computer's Ethernet interface in the “Scan on network interface:” field. Then click the “Scan” button.
Now you can select the generator and change the IP address by right-clicking. This will open the “Slave Addressing” window. Uncheck the ‘DHCP’ box (disable DHCP) to assign the desired IP address “XXX.XXX.XXX.XXX”. After changing the IP address, it is necessary to briefly interrupt the 24 V power supply in order to check the entered data again. If the data has been changed as desired, the program can be closed.

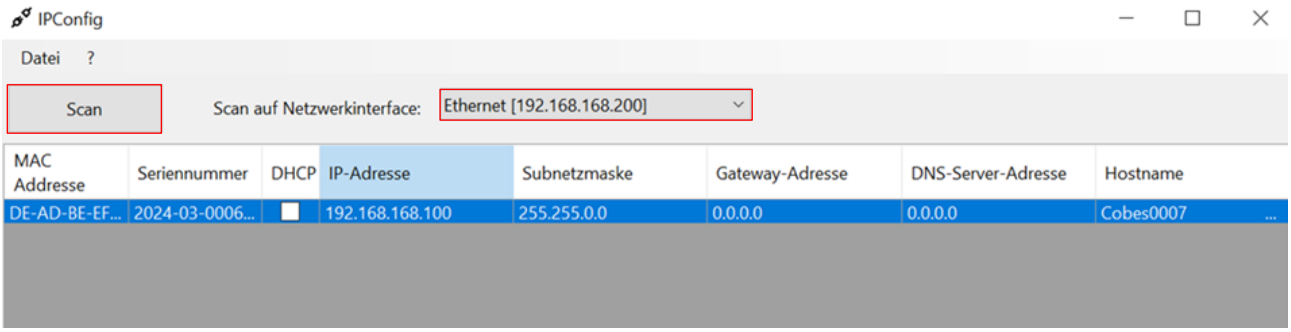


Illustration 9: IPConfig-Tool

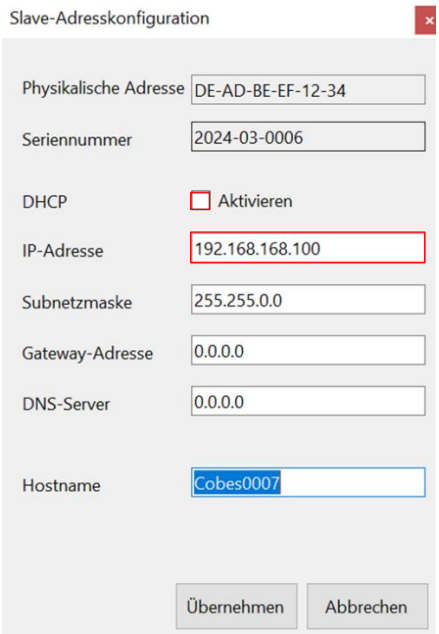


Illustration 10: Slave address configuration

6.5 Status display

Status display Network interface

The status of the network interface is indicated by the two link LEDs.

LED-Status	Status
Off	Network connection disconnected at the physical level
Flashing	Active data exchange
Steady light	Network connection established at the physical level

Processor status display (MCU status)

The status display via the CPU LED can be broken down as shown in the following table.

LED-Color	Function	Slow flashing ≈ 1 Hz; Fast flashing ≈ 2.5 Hz
Green	Steady light	Ready-State
	Fast flashing	Generator in active state (Manual operation or heating program is running)
Yellow	Steady light	Initialization after reset, waiting until network connection has been established for the first time and DC link is charged.
	Fast flashing	Flashing for 10 seconds for system identification in the network.
Red	Steady light	ICC is in error state.
	Slow flashing	Network error (physical connection disconnected or bus master timeout).
	Fast flashing	Shutdown by interlock is active.
Red-Yellow	Fast flashing	Non-acknowledgable error in safety function.

6.6 Switching on the device

Connect the device to the power supply.
Ensure that the mains voltage matches the specifications on the type plate.

- Switch on the cooling water supply for the induction system.
- Supply the generator with mains voltage (400V, 16A and 24V, max. 3.5A).
- The generator is switched on and off via the digital interface.
- The emergency stop must not be activated.
- The generator is ready for operation.

6.7 Switching off the device

The generator is switched on and off via the digital interface.

6.8 Disassembly, packaging

De-energize

- Turn off the generator (for instructions, see section “6.7”).
- Open the disconnect device in the external electrical installation.
- Remove all electrical connections.

Empty the cooling water circuit

Blow out the water circuit with compressed air against the normal flow direction. Remove the water connections on the power supply unit and completely empty the cooling water circuit. To do this, carefully blow out the cooling water circuit with compressed air to remove any water residues.

Packaging

Pack the generator according to its weight and size.

Disposal

Old devices that are no longer in use must be disposed of properly. The generator consists mainly of the following materials:

- Metals
  - Aluminum
  - Copper
  - Brass
  - Galvanized sheet steel

- Plastics
  - Teflon
  - PA and others

7. Working with the device

Danger!

Voltages at the inductor!
When the generator is switched on, the following voltages are present at the inductor:
MF voltage between the two inductor terminals and MF voltage to ground.
➲ The user must install suitable covers and safety devices to protect against contact!

7.1 Operating the device

The device can be controlled via the Ethernet UDP serial interface.
The Ethernet UDP interface is described in Chapter 9.1 ff.

In addition/alternatively, a numerical control (PLC) with touch display can be ordered.
The PLC control is described in Chapter 9.8.

7.2 Error messages

The error messages are described in Chapter 9.8.3 Error Page.

7.3 Maintenance

There are no parts in the device itself that require maintenance.

Check the covers and the function of the safety devices regularly.

Cleaning the water filter
Depending on the quality of the cooling water, the filters in the recooling system may become dirty after prolonged operation. Therefore, all water filters must be cleaned regularly.

General visual inspection
A general visual inspection should be carried out regularly. Check the induction generator for moisture, corrosion, dirt accumulation on the induction coil, and signs of flashover. Check the inductor for mechanical wear.

The connection jaws must be checked regularly for contamination and cleaned if necessary. The areas between and above and below the copper jaws are important; there must be no short-circuit contact here.

Replacing the inductor

It may be necessary to replace the inductor because the original inductor is defective or because a different workpiece geometry requires a new inductor. To do this, proceed in the order described below:

De-energize

- Switch off the generator (for instructions, see section “6.8”).

Drain the cooling water circuit

Blow out the water circuit with compressed air against the normal flow direction. Remove the water connections on the power supply unit and drain the cooling water circuit completely. To do this, carefully blow out the cooling water circuit with compressed air to remove any water residues.

Detach the inductor from the inductor connection jaws

- Loosen the 4 screws on the inductor connection jaws. Remove the inductor.
- Ensure that the two O-rings remain in the grooves of the inductor connection jaws.

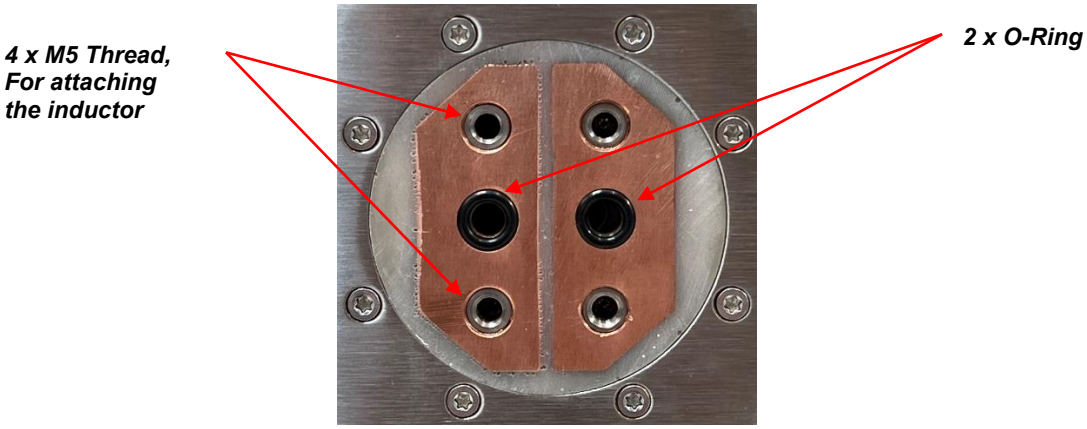


Illustration 11: Inductor connection jaws without inductor

Attaching the new inductor to the inductor connection jaws

- Ensure that both the connection jaws of the inductor and the generator are clean and free of corrosion.
- Ensure that the two O-rings are correctly positioned in the grooves of the inductor connection jaws of the generator.
- Attach the inductor to the inductor connection jaws using the 4 screws provided. Never use conventional steel screws for this purpose!
- Slowly open the cooling water supply and check the connection for leaks.
- Switch the generator back on and slowly increase the setpoint to check the current consumption.

INDUCTION HEATING SYSTEM

8. Technical specifications

Parameters	
Mains connection	3/PE 400V, 50/60Hz
Max. power consumption	5.6kVA
Nominal power	5kW
Max. inductor voltage	130Veff
Recommended backup fuse	Circuit breaker 3x400V, K 16A,
Duty cycle	100%
Cooling water flow rate	1 to 8 l/min (observe cooling water requirements!)
Cooling water pressure (min/max)	3.5-6 bar; max. 8 bar, differential pressure of min. 3.5 bar
Cooling water temperature (max)	40°C
Operating frequency	20 to 100 kHz
Dimensions	75 x 75 x 299 mm (330 mm incl. connections)
Weight	Approx. 4 kg
Protection class	IP 54 with additional cover
Protection rating	I

In general:

	Operation[1]	**Storage**	**Transport**
Temperature	+10 °C - +45 °C	-20 °C - +55 °C	-20 °C - +70 °C
Relative humidity	10 % - 90 %	30 % - 95 %	95 %

1) No condensation or frosting.

9. Appendix

9.1 Quick overview of process data (UDP control)

The telegrams with the PLC output data must be sent cyclically from the bus master to the ICC (port 5010).

9.1.1 PLC output data

Byte#	Description	Data type	Value
1	Prefix	ARRAY [1..8] OF BYTE	B
2			S
3			T
4			S
5			T
6			A
7			R
8			T
9	Control word	UNIT	cf. 9.3.1
10			
11	Current setpoint	REAL	cf. 9.3.2
12			
13			
14			
15	Power setpoint	REAL	cf. 9.3.3
16			
17			
18			
19	Heating program number	UNIT	cf. 9.3.4
20			
21	Suffix	ARRAY [1..8] OF BYTE	B
22			S
23			T
24			E
25			N
26			D
27			T
28			X

Table 1: Byte-wise arrangement of the PLC output data (control data)

9.1.2 SPS-Eingangsdaten

Byte#	Description	Data type	Value
1	Prefix	ARRAY[1..8] OF BYTE	B
2			S
3			T
4			S
5			T
6			A
7			R
8			T
9	Status word	UINT	cf. 9.2.1
10			
11	Process status word	UINT	cf. 9.2.2
12			
13	Actual value Primary current	REAL	cf. 9.2.3
14			
15			
16			
17	Actual value Phase angle	REAL	cf. 9.2.4
18			
19			
20			
21	Actual value Operating frequency	REAL	cf. 9.2.5
22			
23			
24			
25	Actual value Power	REAL	cf. 9.2.6
26			
27			
28			
29	Actual value Capacitor voltage	REAL	cf. 9.2.7
30			
31			
32			
33	Actual value DC link voltage	REAL	cf. 9.2.8
34			
35			
36			
37	Actual value Energy	REAL	cf. 9.2.9
38			
39			
40			
41	Actual value	REAL	cf. 9.2.10
42			
43			
44			
45	Cooling water volume flow	REAL	cf. 9.2.11
46			
47			
48			
49	Actual value Temperature at selected sensor	UINT	cf. 9.2.12
50			
51		UINT	cf. 9.2.13
52			
53	Selected heating program	UINT	cf. 9.2.14
54			
55	Heating program phase Status State machine	DWORD	cf. 9.2.15
56			
57			
58			
59	Status word Error flags	DWORD	cf. 9.2.16
60			
61			
62			
63		ARRAY [1..8] of DWORD	cf. 9.2.17
..			
..			
94			

95	Temperature of the individual heating phases of the current heating program	ARRAY [1..8] of REAL	cf. 9.2.18
..			
..			
126			
127	Suffix	ARRAY[1..8] OF BYTE	B
128			S
129			T
130			E
131			N
132			D
133			T
134			X

Table 2: Byte-wise arrangement of PLC input data (status data)

9.2 Process image PLC input data (ICC → PLC)

The telegrams containing the PLC input data are sent cyclically by the ICC to the bus master (port 5010) at intervals of 100 ms.

9.2.1 Inverter status word – status bit field

The inverter status word contains information on the current operating mode and any errors that have occurred.

Bit 15	Bit 14	Bit 13	Bit 12	Bit 11	Bit 10	Bit 9	Bit 8	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
HEART BEAT	---	---	---	---	---	---	ESTOP	---	---	---	HEAT PROG ERR	HEAT PROG DONE	ERROR	ACTIVE	READY

READY	**status bit „READY“**	
	0	Control system and converter are not ready.
	1	Control system and converter are ready.
ACTIVE	**status bit „ACTIVE“**	
	0	The inverter is not in active mode.
	1	The inverter is in active mode.
ERROR	**status bit „ERROR“**	
	0	There is no fault.
	1	There is at least one fault. The inverter is disabled until the fault has been rectified and acknowledged.
HEAT PROG DONE	**status bit „HEATING PROGRAM DONE“**	
	0	Heating program is running or was not completed successfully
	1	Heating program was completed successfully
HEAT PROG ERR	**status bit „HEATING PROGRAM ERROR“**	
	0	Heating program is running or has been successfully completed
	1	Heating program terminated with error
Bits 5…7	**Reserved**	
ESTOP	**status bit „ESTOP“– Emergency Stop / Hardware Interlock**	
	0	Umrichter befindet sich im regulären Betriebszustand
	1	Die Hardwarefreigabe (Interlock) ist nicht gesetzt. Der Umrichter ist in der Sicherheitsabschaltung.
Bits 9…14	**Reserved**	
HEART BEAT	**status bit „Lifebit“ – Lebenszeichenüberwachung**	
		Toggle bit: The life bit from the control word is returned inverted.

9.2.2 Controller status word – process status bitfield

The controller status word contains more detailed information about the operating mode.

Bit 15	Bit 14	Bit 13	Bit 12	Bit 11	Bit 10	Bit 9	Bit 8	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
---	---	---	---	---	---	---	---	---	---	---	---	CTRL_STATE			

Bits 0...3	**Status display "CTRL_STATE" – Current controller status**	
	0 = (0b0000)	Controller inactive / Power output switched off
	1 (0b0001)	Current limitation
	2 (0b0010)	Power limitation
	3 (0b0011)	Voltage limitation (U_Cap)
	4 (0b0100)	Phase angle limitation φ
	5 (0b0101)	Switching losses limitation PV = UDC · I · f · sin(φ)
	6 (0b0110)	Frequency limitation
	7…15 (0b0111 ⋮ 0b1111)	Reserved.
Bit 4	Violation of minimum limit in heating program	
Bit 5	Violation of maximum limit in heating program	
Bits 6...15	Reserved	

9.2.3 Instantaneous value of primary current

Bit 31	Bit 30	Bit 29	Bit 28	Bit 27	Bit 26	Bit 25	Bit 24	Bit 23	Bit 22	Bit 21	Bit 20	Bit 19	Bit 18	Bit 17	Bit 16	Bit 15	Bit 14	Bit 13	Bit 12	Bit 11	Bit 10	Bit 9	Bit 8	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Instantaneous value of primary current (32-bit floating point number)																															

Instantaneous value Primary current	Primary current of the converter in amperes.

9.2.4 Instantaneous value phase shift primary current

Bit 31	Bit 30	Bit 29	Bit 28	Bit 27	Bit 26	Bit 25	Bit 24	Bit 23	Bit 22	Bit 21	Bit 20	Bit 19	Bit 18	Bit 17	Bit 16	Bit 15	Bit 14	Bit 13	Bit 12	Bit 11	Bit 10	Bit 9	Bit 8	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Instantaneous value of welding voltage (32-bit floating point number)																															

Instantaneous value Phase shift	Phase shift in degrees

9.2.5 Instantaneous value frequency

Bit 31	Bit 30	Bit 29	Bit 28	Bit 27	Bit 26	Bit 25	Bit 24	Bit 23	Bit 22	Bit 21	Bit 20	Bit 19	Bit 18	Bit 17	Bit 16	Bit 15	Bit 14	Bit 13	Bit 12	Bit 11	Bit 10	Bit 9	Bit 8	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Instantaneous value frequency (32-bit floating point number)																															

Instantaneous value frequency	Operating frequency in Hertz (switching frequency of the inverter, setpoint of the modulator)

9.2.6 Instantaneous power value

Bit 31	Bit 30	Bit 29	Bit 28	Bit 27	Bit 26	Bit 25	Bit 24	Bit 23	Bit 22	Bit 21	Bit 20	Bit 19	Bit 18	Bit 17	Bit 16	Bit 15	Bit 14	Bit 13	Bit 12	Bit 11	Bit 10	Bit 9	Bit 8	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Instantaneous power value (32-bit floating point number)																															

Instantaneous power value	Power output in watts, measured in the DC link

9.2.7 Instantaneous value of capacitor voltage

Bit 31	Bit 30	Bit 29	Bit 28	Bit 27	Bit 26	Bit 25	Bit 24	Bit 23	Bit 22	Bit 21	Bit 20	Bit 19	Bit 18	Bit 17	Bit 16	Bit 15	Bit 14	Bit 13	Bit 12	Bit 11	Bit 10	Bit 9	Bit 8	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Instantaneous value of capacitor voltage (32-bit floating point number)																															

Instantaneous value of capacitor voltage	Calculated capacitor voltage in volts

9.2.8 Instantaneous value of DC link voltage

Bit 31	Bit 30	Bit 29	Bit 28	Bit 27	Bit 26	Bit 25	Bit 24	Bit 23	Bit 22	Bit 21	Bit 20	Bit 19	Bit 18	Bit 17	Bit 16	Bit 15	Bit 14	Bit 13	Bit 12	Bit 11	Bit 10	Bit 9	Bit 8	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Instantaneous value of DC link voltage (32-bit floating point number)																															

Instantaneous value DCL voltage	DC link voltage in volts

9.2.9 Instantaneous value of energy meter

Bit 31	Bit 30	Bit 29	Bit 28	Bit 27	Bit 26	Bit 25	Bit 24	Bit 23	Bit 22	Bit 21	Bit 20	Bit 19	Bit 18	Bit 17	Bit 16	Bit 15	Bit 14	Bit 13	Bit 12	Bit 11	Bit 10	Bit 9	Bit 8	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Instantaneous value of energy meter (32-bit floating point number)																															

Instantaneous value Energy	Energy meter in watt seconds

9.2.10 Instantaneous value of cooling water flow rate

Bit 31	Bit 30	Bit 29	Bit 28	Bit 27	Bit 26	Bit 25	Bit 24	Bit 23	Bit 22	Bit 21	Bit 20	Bit 19	Bit 18	Bit 17	Bit 16	Bit 15	Bit 14	Bit 13	Bit 12	Bit 11	Bit 10	Bit 9	Bit 8	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Instantaneous value of cooling water flow (32-bit floating point number)																															

Instantaneous value of cooling water flow	Measured value of the volume flow in l/min

9.2.11 Instantaneous value of temperature sensor

Bit 31	Bit 30	Bit 29	Bit 28	Bit 27	Bit 26	Bit 25	Bit 24	Bit 23	Bit 22	Bit 21	Bit 20	Bit 19	Bit 18	Bit 17	Bit 16	Bit 15	Bit 14	Bit 13	Bit 12	Bit 11	Bit 10	Bit 9	Bit 8	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Instantaneous value of temperature sensor (32-bit floating point number)																															

Instantaneous value of temperature sensor	Measured value of the selected temperature sensor in °C

9.2.12 Instantaneous value of heating program phase

Bit 15	Bit 14	Bit 13	Bit 12	Bit 11	Bit 10	Bit 9	Bit 8	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Instantaneous value of heating program phase (16-Bit Unsigned Integer)															

Instantaneous value of heating program phase	Number of the active phase of the selected heating program, 0 if heating program is not active

9.2.13 Current value of heating program phase

Bit 15	Bit 14	Bit 13	Bit 12	Bit 11	Bit 10	Bit 9	Bit 8	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Current value of the heating program phase currently being (16-Bit Unsigned Integer)															

Instantaneous value Heating program phase	Number of the active phase of the selected heating program, 0 if heating program is not active

9.2.14 Status state machine – state

This 16-bit number indicates the current status of the converter's primary control logic.

Bit 15	Bit 14	Bit 13	Bit 12	Bit 11	Bit 10	Bit 9	Bit 8	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
FSM Status (16-Bit Unsigned Integer)															

	Status state machine	
FSM Status	0	Reserved: Required for internal use [NONE]
	1	Reserved: Required for internal use [_ANY]
	2	Init: The drive is in the start procedure; the internal components are being activated and prepared for operation.
	3	InitNetwork: Start and initialize the network interface until a physical connection is established and a valid IP address configuration has been set (either via DHCP or static address).
	4	StartNetServices: Start network services (fieldbus communication, service protocol, and others, if applicable).
	5	WaitForBusMaster: Wait until a communication connection with the higher-level controller has been established.
	6	PrepareOnBoardSupplies: Activate the internally generated supply voltages.
	7	PrepareGateDrivers: Initialize and check the gate drivers.
	8	WaitForDCLinkCharged: Wait until the DC link is charged before activating the precharge bypass.
	9	Ready: Ready state.
	10	Active: Active operating mode with enabled power output.
	11	Error: Error state. Can only be exited by acknowledging.
	12	NetworkError: A network error has occurred, loss of bus master communication (timeout), physical connection disconnected, or other network errors.
	13	
	14	UnrecoverableError: A serious, non-acknowledgeable error has occurred. The system must be restarted.

9.2.15 Status word error

Bit 31	Bit 30	Bit 29	Bit 28	Bit 27	Bit 26	Bit 25	Bit 24	Bit 23	Bit 22	Bit 21	Bit 20	Bit 19	Bit 18	Bit 17	Bit 16	Bit 15	Bit 14	Bit 13	Bit 12	Bit 11	Bit 10	Bit 9	Bit 8	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Instantaneous value Error status word (32-Bit Unsigned Integer)																															

Instantaneous value Error status word	The possible system errors are mapped to the bits:

Bit	Triggering error
0	Error in PWM measurement evaluation (phase detection, galvanically isolated measured values)
1	Error in pulse width modulator
2	Gate driver error
3	Overcurrent primary current
4	Overcurrent DC link
5	Overvoltage DC link
6	Undervoltage DC link
7	Capacitive switching: tolerance limit exceeded
8	Overtemperature: Transformer temperature above limit value
9	Overtemperature: Heat sink temperature above set limit value
10	Overtemperature: Pyrometer temperature above set limit value
11	Overtemperature: Thermocouple temperature above set limit value
12	Coolant flow monitor: Minimum volume flow below limit value
13	Reserved.
14	Sensor cable interrupted or sensor not plugged in
15	Sensor data implausible
16	Supply voltage: External 24 volt voltage limit values violated
17	Supply voltage: Internal 15 V voltage limit values violated
18	Supply voltage: Internal 5 V voltage limit values violated
19	Supply voltage: Internal 3.3 V voltage (digital) limit values violated
20	Supply voltage: Internal 3.0 V voltage (analog) limit values violated
21	Supply voltage: Internal isolated power supply limit values violated
22	Supply voltage: Gate driver power supply limit values violated
23	Reserved.
24	Reserved.
25	Safety function: Locking of safety-critical functions could not be verified
26	Safety function: Power enabled blocked because interlock was triggered
27	Communication error (life bit faulty)
28	Communication error (bus master timeout)
29	Memory error: Stack overflow
30	Program error / Processed data invalid
31	Undefined error in program or communication

9.2.16 Status word error current heating program

Bit 31	Bit 30	Bit 29	Bit 28	Bit 27	Bit 26	Bit 25	Bit 24	Bit 23	Bit 22	Bit 21	Bit 20	Bit 19	Bit 18	Bit 17	Bit 16	Bit 15	Bit 14	Bit 13	Bit 12	Bit 11	Bit 10	Bit 9	Bit 8	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Instantaneous value Error status word (32-Bit Unsigned Integer)																															

Instantaneous value Error status word	Die möglichen Systemfehler werden auf die Bits abgebildet:

Bit	Triggering error
0	Reserved
1	Invalid temperature sensor during a heating phase
2	No temperature sensor during a heating phase
3-7	Reserved
8	Temperature above limit value
9	Minimum temperature value not reached
10	Energy above limit value
11	Minimum energy value not reached
12-15	Reserved
16-23	Heating phase that triggered the error
24-31	Current heating program that triggered the error

9.2.17 Energy meter for heating phases for current heating program

Bit 255	Bit 30	Bit 29	Bit 28	Bit 27	Bit 26	Bit 25	Bit 24	Bit 23	Bit 22	Bit 21	Bit 20	Bit 19	Bit 18	Bit 17	Bit 16	Bit 15	Bit 14	Bit 13	Bit 12	Bit 11	Bit 10	Bit 9	Bit 8	Bit 7	Bit 6	Bit 5	Bit 32	Bit 31	Bit 2	Bit 1	Bit 0
Bit 255 … Bit 224				Bit 223 … Bit 192				Bit 191 … Bit 160				Bit 159 … Bit 128				Bit 127 … Bit 96				Bit 95 … Bit 64				Bit 63 … Bit 32				Bit 31 … Bit 0			
energy meter Phase 8 (32-Bit UINT)				energy meter Phase 7 (32-Bit UINT)				energy meter Phase 6 (32-Bit UINT)				energy meter Phase 5 (32-Bit UINT)				energy meter Phase 4 (32-Bit UINT)				energy meter Phase 3 (32-Bit UINT)				energy meter Phase 2 (32-Bit UINT)				energy meter Phase 1 (32-Bit UINT)			

energy meter	Energy meter in watt seconds

9.2.18 Temperature value of the heating phases for the current heating program

Bit 255	Bit 30	Bit 29	Bit 28	Bit 27	Bit 26	Bit 25	Bit 24	Bit 23	Bit 22	Bit 21	Bit 20	Bit 19	Bit 18	Bit 17	Bit 16	Bit 15	Bit 14	Bit 13	Bit 12	Bit 11	Bit 10	Bit 9	Bit 8	Bit 7	Bit 6	Bit 5	Bit 32	Bit 31	Bit 2	Bit 1	Bit 0
Bit 255 … Bit 224				Bit 223 … Bit 192				Bit 191 … Bit 160				Bit 159 … Bit 128				Bit 127 … Bit 96				Bit 95 … Bit 64				Bit 63 … Bit 32				Bit 31 … Bit 0			
final temperature Phase 8 (32-Bit UINT)				final temperature Phase 7 (32-Bit UINT)				final temperature Phase 6 (32-Bit UINT)				final temperature Phase 5 (32-Bit UINT)				final temperature Phase 4 (32-Bit UINT)				final temperature Phase 3 (32-Bit UINT)				final temperature Phase 2 (32-Bit UINT)				final temperature Phase 1 (32-Bit REAL			

final temperature	Measured value of the selected temperature sensor in °C

9.3 Process image PLC output data (PLC → ICC)

9.3.1 Control word converter – control bitfield

The converter control word supplies the converter with all control signals required for operation.

Bit 15	Bit 14	Bit 13	Bit 12	Bit 11	Bit 10	Bit 9	Bit 8	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
HEART BEAT	---	---	---	---	---	---	ACK ERROR	RESET ENRGY	---	---	---	SEL TEMP		CTRL MODE	HEIZEN EIN

HEATING ON	**Control bit "HEATING ON" – Enable control in manual mode**	
	0	Switch off the inverter
	1	Switch on the inverter with the selected controller mode
CTRL MODE	**Control bit "CONTROL MODE" – Switching operating mode**	
	0	Manual operation (setpoint specification via PLC)
	1	Automatic operation (heating programs)
SEL TEMP	**Control bits "SEL TEMP SOURCE" – Switch between thermocouple or pyrometer**	
	00	No temperature sensor selected. Temperature monitoring is not taking place.
	01	Monitoring of the thermocouple input. The thermocouple value is used for evaluation for heating programs.
	10	Monitoring of the pyrometer input. The pyrometer value is used for evaluation for heating programs.
	11	*Reserved*
Bits 4..6	**Reserved**	
RESET ENRGY	**Control bit "Reset Energy" – Reset energy meter**	
	0	Energy meter is enabled and integrates the power output over time
	1	The energy meter remains reset to 0 as long as the bit is set
ACK ERR	**Control bit "Acknowledge Error" – Acknowledge error**	
	0	No action.
	0 → 1	When the edge is positive, a pending error is acknowledged and the inverter exits the error state.
Bits 9..14	**Reserved**	
Bit 15	**Heart Beat** – Life sign monitoring. The bit state must change every second.	

9.3.2 Target current setting – target current (input)

Bit 31	Bit 30	Bit 29	Bit 28	Bit 27	Bit 26	Bit 25	Bit 24	Bit 23	Bit 22	Bit 21	Bit 20	Bit 19	Bit 18	Bit 17	Bit 16	Bit 15	Bit 14	Bit 13	Bit 12	Bit 11	Bit 10	Bit 9	Bit 8	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Current setpoint (32-bit floating point number)																															

Setpoint Current	Setpoint specification in percent, relative to SYS_I_MAX. Replacement control with the power setpoint, i.e., current control only below the set power specification (target power)

9.3.3 Power setpoint specification – target power (input)

Bit 31	Bit 30	Bit 29	Bit 28	Bit 27	Bit 26	Bit 25	Bit 24	Bit 23	Bit 22	Bit 21	Bit 20	Bit 19	Bit 18	Bit 17	Bit 16	Bit 15	Bit 14	Bit 13	Bit 12	Bit 11	Bit 10	Bit 9	Bit 8	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Target value power (32-bit floating point number)																															

Target value power	Setpoint specification in percent, relative to SYS_P_MAX. Replacement control with the current setpoint, i.e., power control only below the set current setpoint.

9.3.4 Setpoint specification for heating program

Bit 15	Bit 14	Bit 13	Bit 12	Bit 11	Bit 10	Bit 9	Bit 8	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
Heating program setpoint (16-bit Unsigned Integer)															

Setpoint Heating program no.	Setpoint specification Heating program number [1...100]

9.4 Service communication

The ICC controller implements a dedicated service task that opens a UDP socket on port 4660. Service telegrams for the bidirectional exchange of parameters, diagnostic data, and heating programs are exchanged on this port.

In addition, the ICC sends and receives UDP datagrams on port 4661, which contain text output with information on program execution, status and error messages. This console output can be used for troubleshooting, among other things. A console session must be initiated with the “HELLO” command before the output is displayed.

The ICC sends the response telegrams for service telegrams and console output back to the port from which the datagrams were sent.

9.5 Heating programs

The ICC control can use and store 100 heating programs. A heating program is described by eight heating phases. The higher-level PLC transmits a heating program in a single telegram containing all 8 heating phases.

The heating programs are transmitted by the telegrams described below.

9.5.1 ransferring a heating program

Bytes	Data type	Designation	Description
0 ⋮ 11	CHAR[12]	Preamble	String with “SET_HEATPROG” (No null termination)
12 13	UINT16	HeatProgNumber *N*	Program number N of the program to be written. The first program has N = 1
14 ⋮ 53	HEAT_PHASE_T	HeatProgPhase *N.1*	Data structure of a heating program, see section 9.6
54 ⋮ 93	HEAT_PHASE_T	HeatProgPhase *N.2*	
94 ⋮ 133	HEAT_PHASE_T	HeatProgPhase *N.3*	
134 ⋮ 173	HEAT_PHASE_T	HeatProgPhase *N.4*	
174 ⋮ 213	HEAT_PHASE_T	HeatProgPhase *N.5*	
214 ⋮ 253	HEAT_PHASE_T	HeatProgPhase *N.6*	
254 ⋮ 293	HEAT_PHASE_T	HeatProgPhase *N.7*	
294 ⋮ 333	HEAT_PHASE_T	HeatProgPhase *N.8*	
334 ⋮ 346	CHAR[12]	Suffix	String with “END_HEATPROG” (No null termination))

9.5.2 Reading a heating program

To read out a heating program, the following telegram must be sent to the ICC

Bytes	Datentyp	Bezeichnung	Beschreibung
1 ⋮ 12	CHAR[12]	Preamble	String with “GET_HEATPROG” (no null termination)
13	UINT16	HeatProgNumber *N*	Program number N of the program to be read
14			
15 ⋮ 26	CHAR[12]	Suffix	String with “END_HEATPROG” (no null termination)

The converter's response telegram has the same structure as described in section 9.5.1, but the prefix then contains “GET_HEATPROG“.

9.6 Establishing a heating phase

A heating phase is 40 bytes in size and has the following data structure:

Bytes	Data type	Designation	Description
1	Bool (≙ UINT8)	Mode	Operating mode 0 = Energy control mode 1 = Temperature control mode
2	UINT8	Forwarding	0 = No switching, waiting for time to elapse 1 = Switching when energy or temperature is reached
3	UINT8	Control mode	0 = Control based on current 1 = Control based on power
4	Bool (≙ UINT8)	Status	0 = Heating phase active 1 = Heating phase inactive
5	UINT32	Current	Current setpoint, in per mille relative to FULL_SCALE_CURRENT
6			
7			
8			
9	UINT32	Power	Power setpoint, in per mille relative to FULL_SCALE_POWER
10			
11			
12			
13	UINT32	Time	Maximum duration of the heating phase Unit: milliseconds [ms]
14			
15			
16			
17	UINT32	Energy setpoint	Setpoint for energy to be supplied in this heating phase Unit: watt seconds [Ws]
18			
19			
20			
21	UINT32	Energy min	Minimum energy to be supplied in this heating phase (lower limit tolerance band test) Unit: watt seconds [Ws]
22			
23			
24			
25	UINT32	Energy max	Maximum energy to be supplied in this heating phase (upper limit tolerance band test) Unit: watt seconds [Ws]
26			
27			
28			
29	UINT32	Temperature setpoint	Target temperature to be reached in this heating phase Unit: degrees Celsius [°C]
30			
31			
32			
33	UINT32	Temperature min	Minimum temperature to be reached in this heating phase (lower limit tolerance band test) Unit: degrees Celsius [°C]
34			
35			
36			
37	UINT32	Temperature max	Maximum temperature to be reached in this heating phase (upper limit tolerance band test) Unit: degrees Celsius [°C]
38			
39			
40			

9.7 Monitoring functions

If the fieldbus fails, the system is switched off and all active operating states are switched to standby mode. Monitoring is performed by two mechanisms:

- Lifebit monitoring (fast, timeout 1 s)
- Telegram monitoring (slow, timeout 10 s)

Both errors put the system into a safe state and must be acknowledged.
The error acknowledgment must then be performed via the field bus, which is running again; at least two successfully processed telegrams are required to detect the signal edges in the control bits.

9.7.1 Missing master life bit

If no change in the life bit of the higher-level controller is detected for longer than 1000 milliseconds, the system switches off and enters the error state.

9.7.2 Timeout due to absence of master telegrams

If no valid telegram from the fieldbus master is detected for longer than 10 seconds, the system switches off and enters the error state.

9.8 PLC control

After starting the control system, it first attempts to establish a connection to the generator. Once the connection is established, the main menu opens. If the connection cannot be established, the “Configuration” settings page opens automatically.

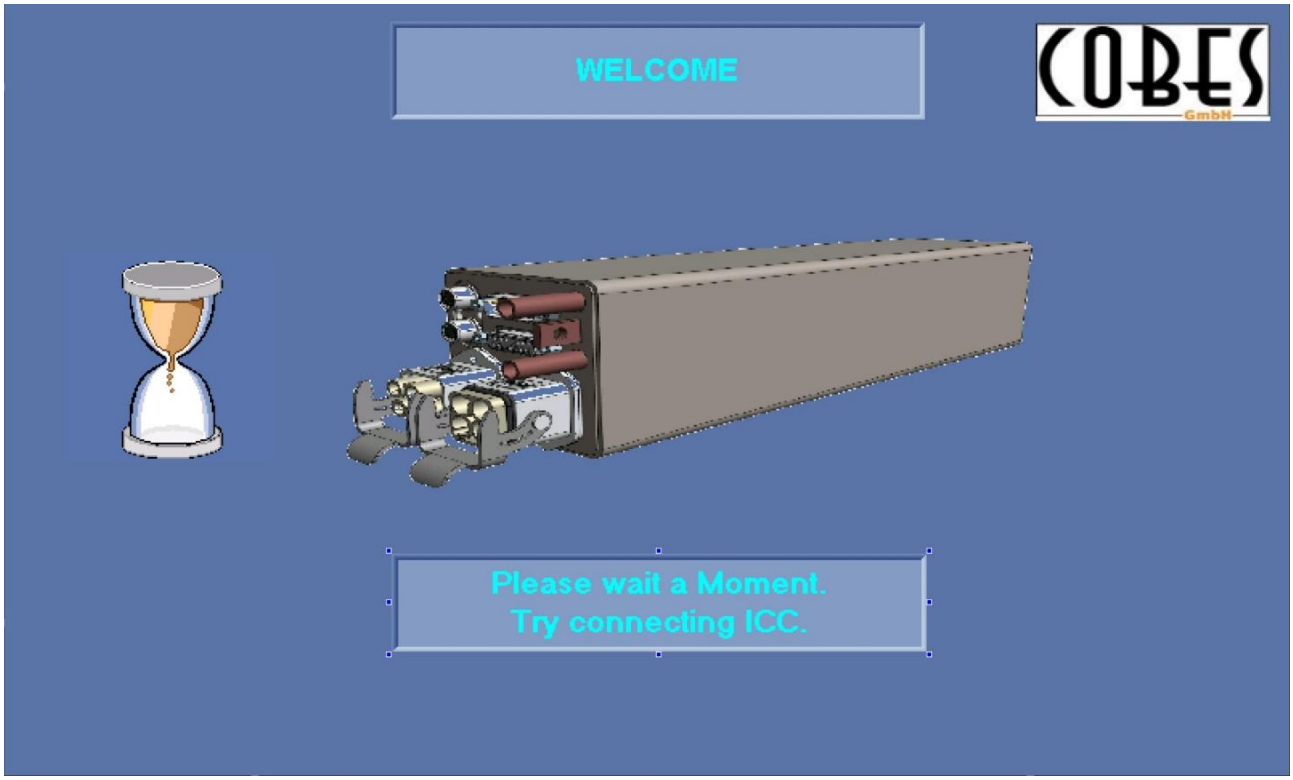


***Illustration* 12: Loading screen for PLC control**

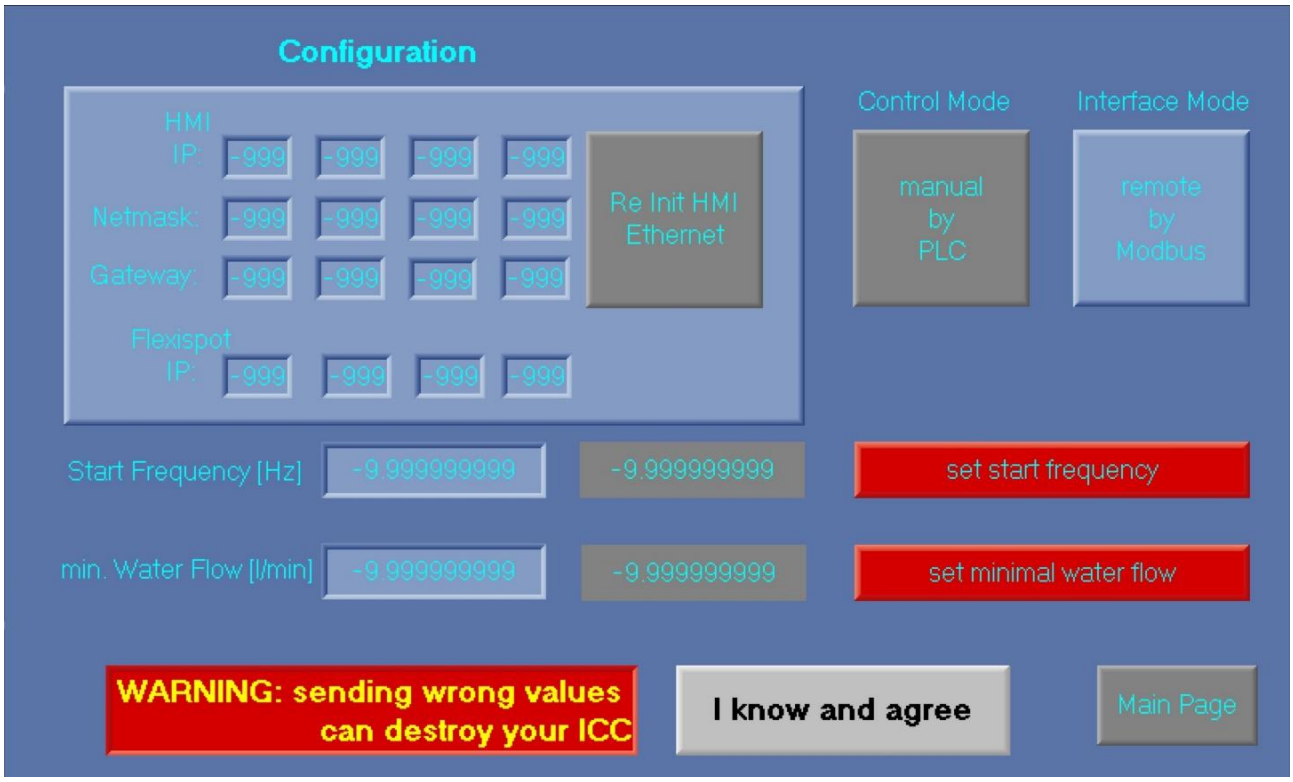


***Illustration* 13: "Configuration" PLC control**

The following changes can be made on the settings page:

Field	Description
HMI-IP	The IP address of the control system is set here
Netmask	Address set
Gateway	Address set
Flexispot IP	The IP address of the ICC generator is set here
Start Frequency [Hz]	The start frequency of the ICC generator is set here Confirm that incorrect values can destroy the generator! The values can be changed in the following window
min. Water Flow [l/min]	The minimum cooling water flow is set here Confirm that incorrect values can destroy the generator! The values can be changed in the following window
Control Mode	Here you can select whether the generator is controlled manually, via the setpoint values, or via the heating programs
Interface Mode	Here you can select whether the input is made via the local HMI or via an external control

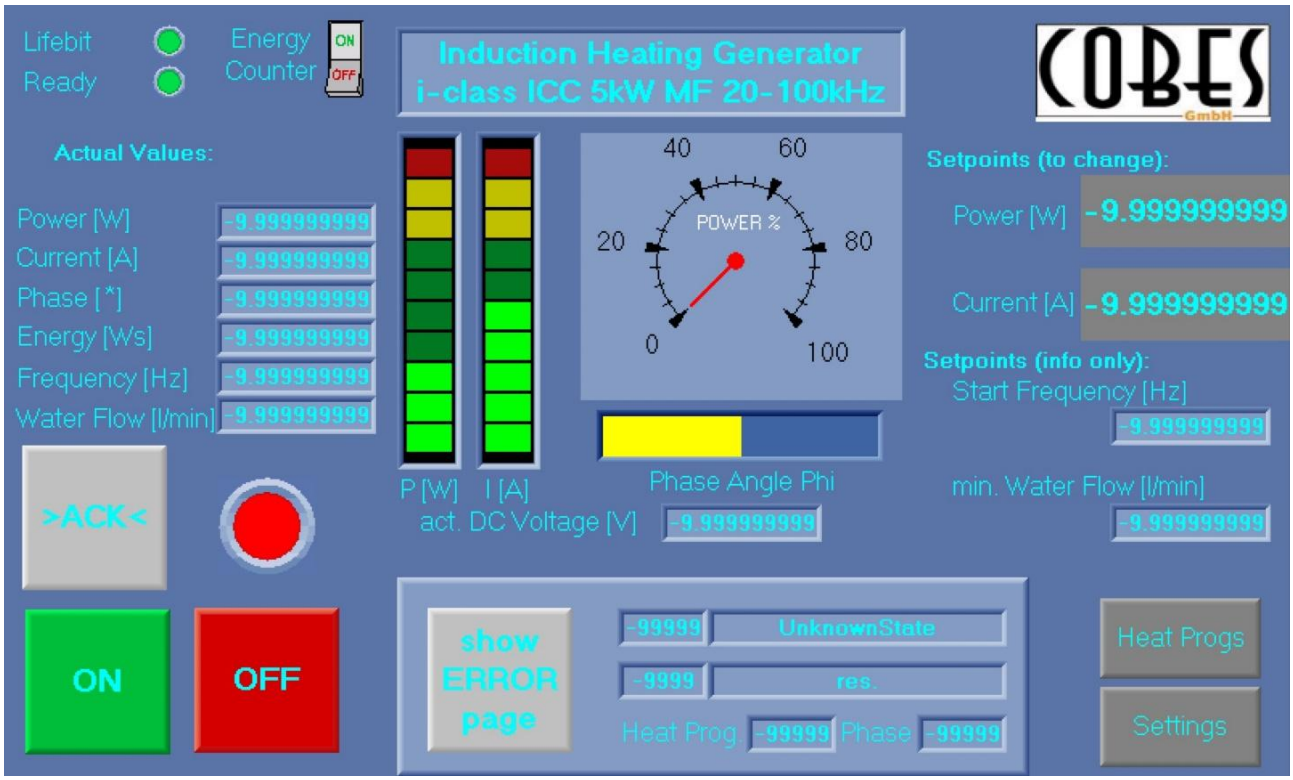


***Illustration* 14: Home screen PLC control**

Field	Description
Lifebit	The LED indicates communication between the generator and the control unit. If this LED is not flashing, there is no communication between the devices.
Ready	The LED indicates the operating status of the generator. Green: Generator is ready for operation. Red: An error has occurred.
Energy Counter	The maximum energy meter is started here. This causes the generator to switch off automatically after a certain value has been reached. This is only possible via a heating program.
Actual Values	Shows the current parameters of the generator (power, current, phase angle, energy conversion, frequency, and water flow).
P [W]; I [A]	Shows the current value for power and current in relation to the maximum values.
Circular diagram for performance	Shows the current power as a percentage.
Phase Angle Phi	Shows the current phase angle.
Act. DC Voltage [V]	Shows the current DC link voltage.
Setpoints (to change)	Input of the setpoints for power and current.
Setpoints (info only)	Shows the current start frequency in Hz and the minimum water flow in l/min.

Field	Description
ON Button / OFF Button	The generator is switched on using the “ON” button. After switching on, the “OFF” button appears so that the generator can be switched off again.
>ACK<	This button appears when there is an error. Once the error has been rectified, it can be acknowledged by pressing and holding the button. The button then disappears again.
Show Error page	This button is used to call up the current error list. It also shows which error bit is being addressed. The limit is displayed below. Heat Prog shows which heating program is currently running. Phase shows which step the program is currently in.
Heat Progs	This menu item takes you to the heating programs.
Settings	This menu item takes you to the generator settings.

9.8.3 Error Page PLC Control

This page displays all errors encountered by the generator.

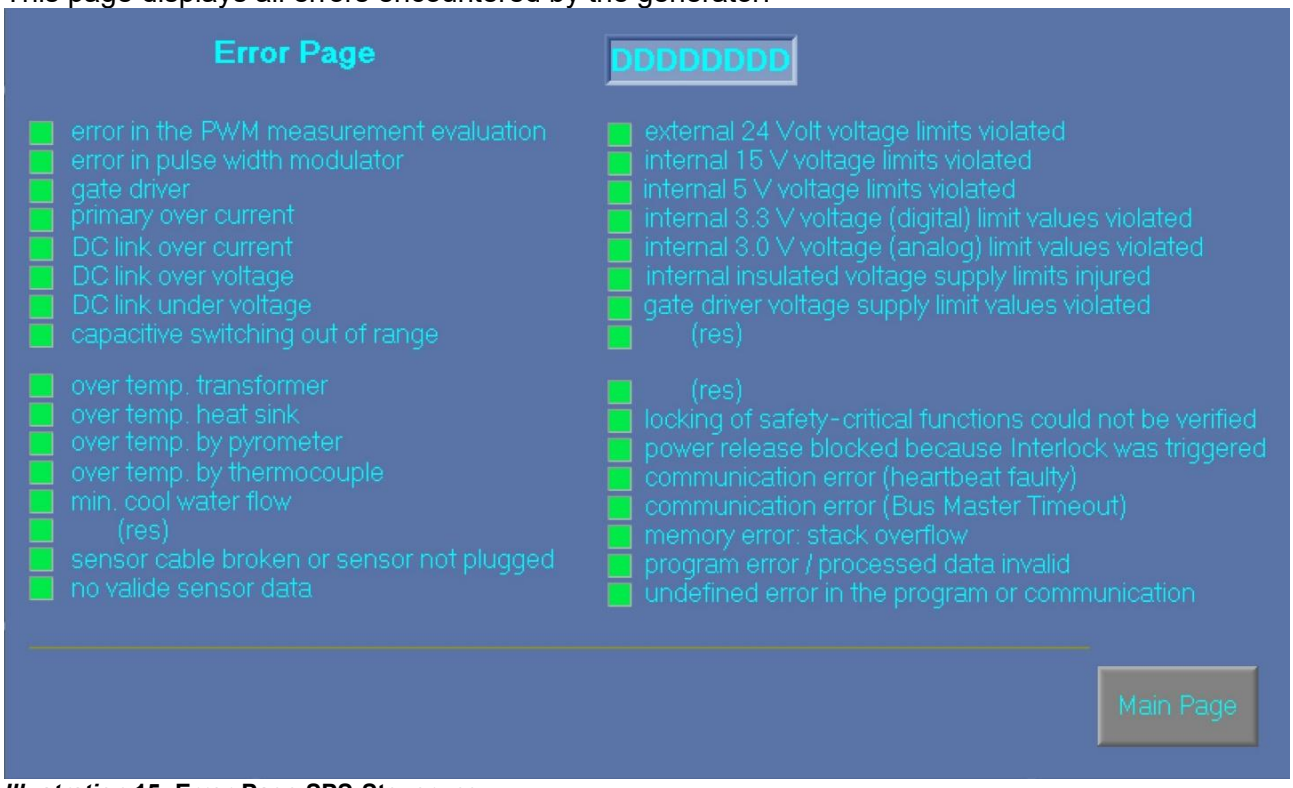


***Illustration* 15: Error-Page SPS-Steuerung**

Field	Description
Green	No error on the generator.
Red	The displayed error is active.
Reset	Errors can be reset in the main menu [>ACK<].
Main Page	This button takes you to the main menu

Error	Solution
Error in the PWM measurement evaluation	Take the generator out of service and send it to COBES.
Error in pulse width modulator	Take the generator out of service and send it to COBES.
Gate driver	Take the generator out of service and send it to COBES.
Primary over current	Adjustment incorrect, inform COBES.
DC link over current	Adjustment incorrect, inform COBES.
DC link over voltage	Check power supply, inform COBES.
Capacitive switching out of range	Adjustment incorrect, inform COBES.

Error	Solution
Over temp. transformer	Take the generator out of service and send it to COBES.
Over temp. heat sink	Take the generator out of service and send it to COBES.
Over temp. by pyrometer	Adjust the timer program.
Over temp. by thermocouple	Adjust the timer program.
min. cool water flow	Check the cooling water flow.
(res)	-
Sensor cable broken or sensor not plugged	Check the sensor.
No valide sensor data	Check the sensor.

Error	Solution
External 24 Volt voltage limits violated	Check external 24-volt power supply.
Internal 15 V voltage limits violated	Take generator out of service and send it to COBES.
Internal 5 V voltage limits violated	Take generator out of service and send it to COBES.
Internal 3.3 V voltage (digital) limit values violated	Take generator out of service and send it to COBES.
Internal 3.0 V voltage supply limits injured	Take generator out of service and send it to COBES.
Gate driver voltage supply limit values violated	Take generator out of service and send it to COBES.
(res)	-

Error	Solution
(res)	
Locking of safety-critical functions could not be verified	Take the generator out of service and send it to COBES.
Power release blocked because Interlock was triggered	Check the interlock circuit.
Communication error (heartbeat faulty)	Check the network cable and IP address.
Communication error (Bus Master Timeout)	Check the network cable and IP address.
Memory error: stack overflow	Take the generator out of service and send it to COBES.
Program error / processed data invalid	Check the generator control system.
Undefined error in the program or communication	Take the generator out of service and send it to COBES.

9.8.4 Heating programs

In the “Heating Program,” 100 heating programs can be created, each with 8 phases.

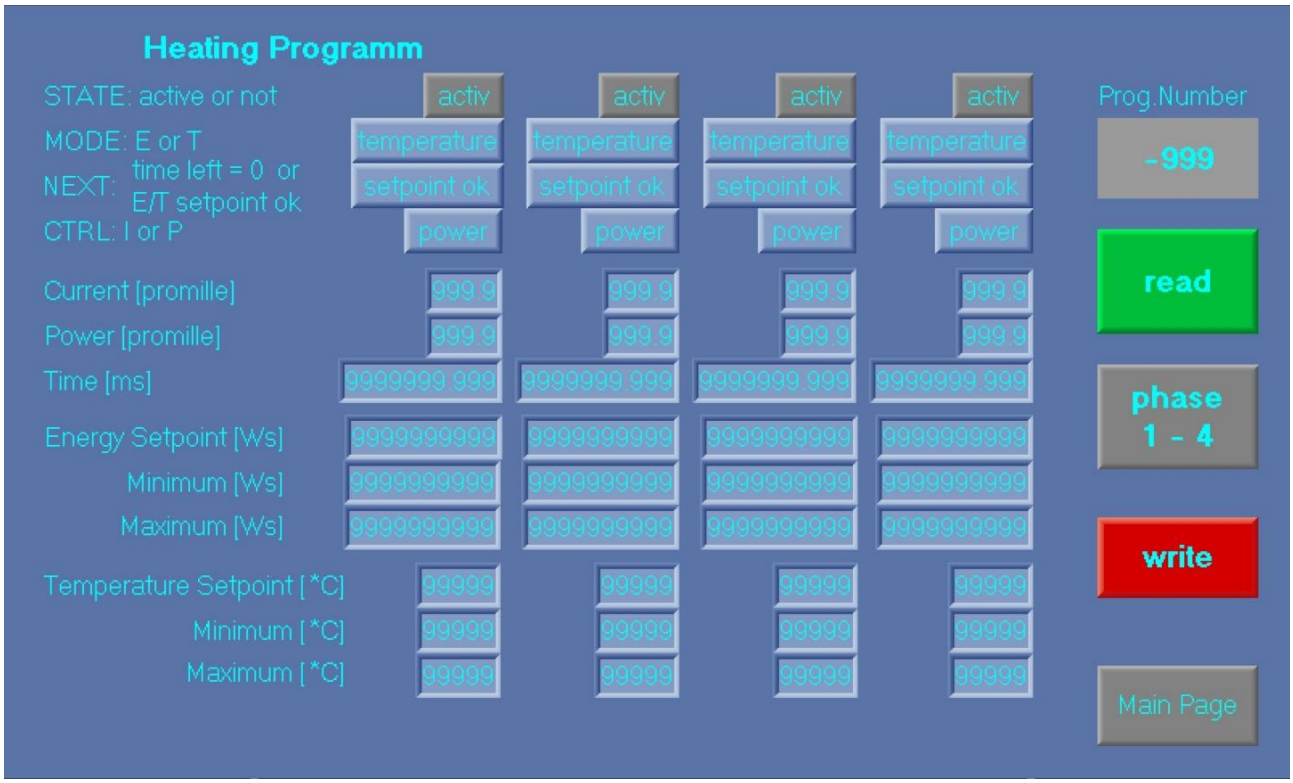


***Illustration* 16: Heating program PLC control**

Field	Description
STATE	Here you can specify whether the program is active or inactive.
MODE	Here you can choose between energy or time.
NEXT	Here you can choose whether the next step starts after a certain time or after a set point is reached.
CTRL	Here you can select whether to set up current or power control.
Current	Here you can set the parameter for the current.
Power	Here you can set the parameter for the power.
Time	Here you can set the parameter for the time.
Energy Setpoint	Here you can enter the energy setpoints.
Temperature Setpoint	Here you can enter the temperature setpoints.
Prog.Number	You can create 100 heating programs, each with 8 phases.
read	Use this button to transfer the generator's heating programs to the control system.
Phase 1-4	Use this button to change the phases of the program. Phase 1-4 or Phase 5-8
write	Use this button to transfer the written heating program to the generator.
Main Page	Use this button to return to the main menu.

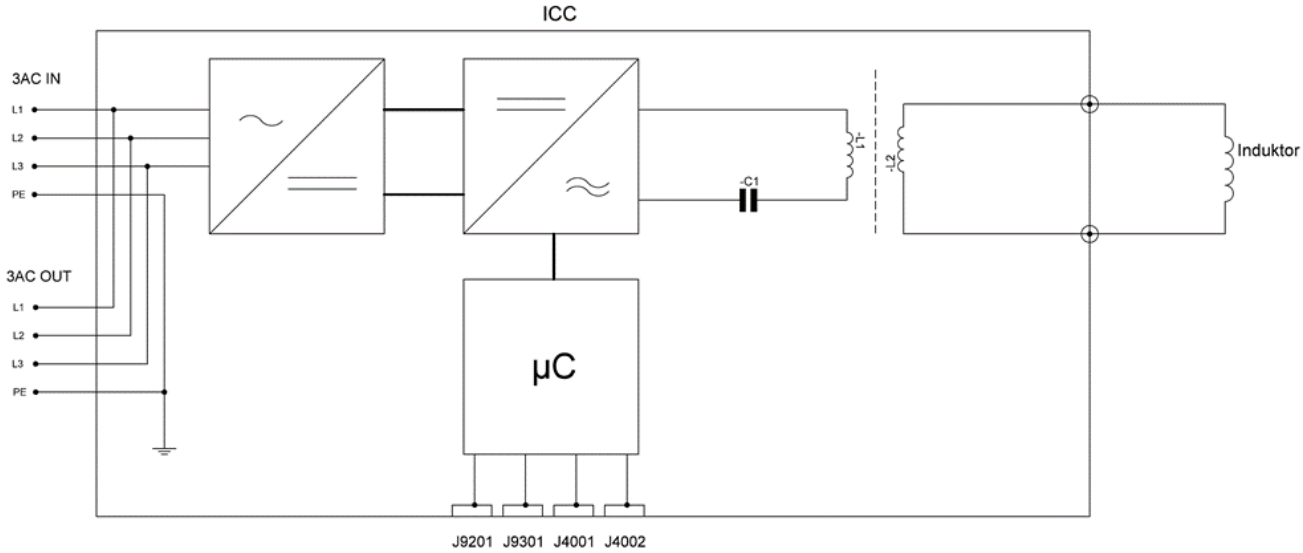


Illustration 17: Overview of basic unit

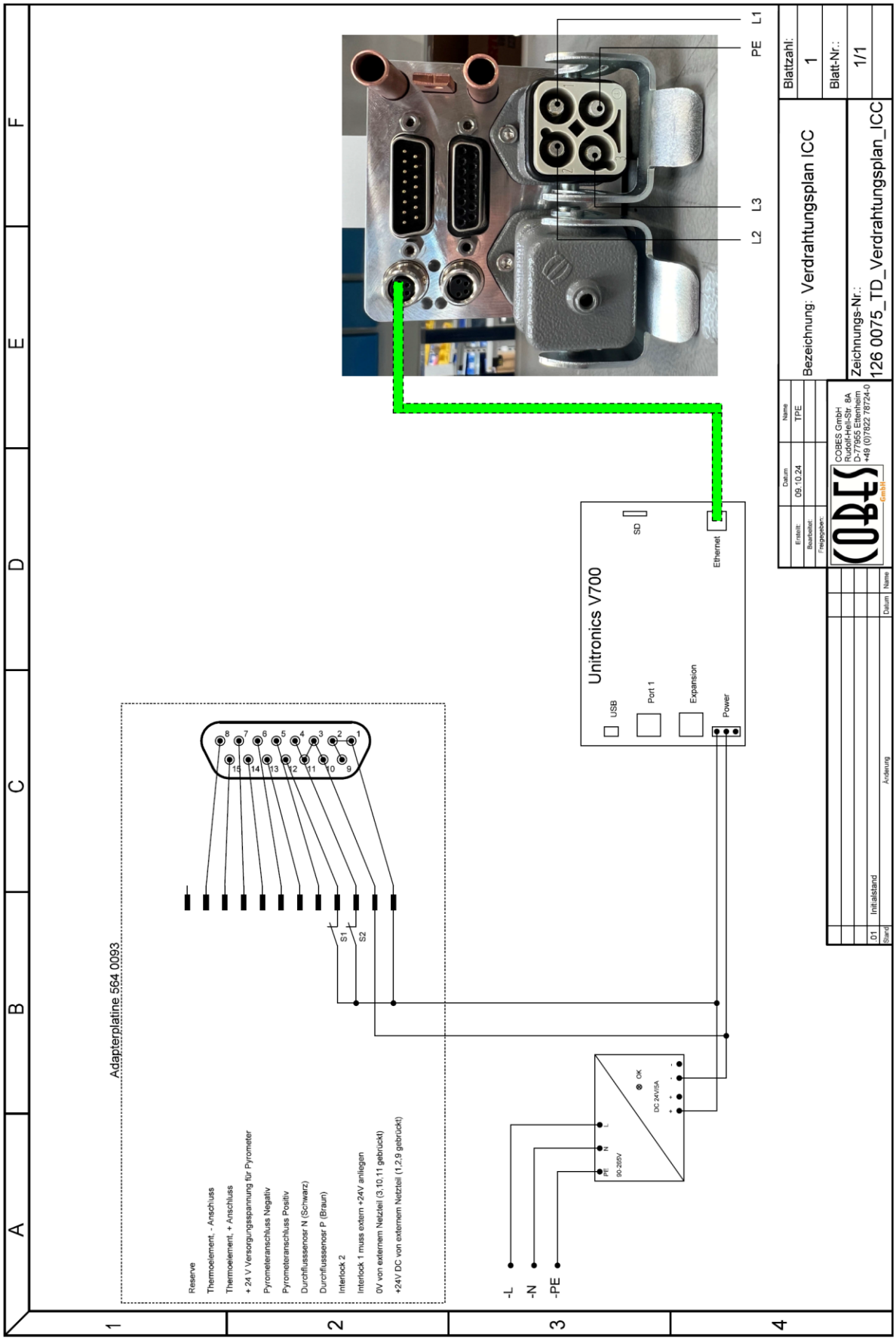


Illustration 18: wiring diagram ICC

9.10 Cooling water requirements

Validity

This specification is valid for a closed system with the cooling water conductivity range specified below. Please refer to the technical data for the generator for the minimum required flow rate. The following applies in general:

- Conductivity range 50 - 600 µS/cm
- Differential pressure approx. 3.5 bar (depending on inductor), maximum inlet pressure 8 bar
- Free of suspended substances
- Total hardness max. 8 °dH
- pH > 7

Definition of a closed system

A cooling circuit is referred to as a “closed system” if the cooling water circulates in a closed circuit and therefore does not come into contact with the surrounding atmosphere or with foreign water from other cooling circuits. This means that it is not possible for the water parameters to be altered by the atmosphere or by foreign water.

Requirements for a functioning cooling circuit

- Dirt traps with a fine mesh of 0.25 mm must be installed in the cooling water supply.
- Do not use transparent hose connections for open installation, as light exposure promotes algae growth.
- TIP: Use opaque covers to protect transparent hoses from light exposure.
- The closed system must not be connected to external cooling circuits.
- The following materials must not come into contact with the system water:
  - Aluminum
  - Carbon steel
  - Galvanized and hot-dip galvanized steel.

Water parameters

TOTAL HARDNESS	MAX. PH VALUE
8 °dH	7,8
6 °dH	8,1
4 °dH	8,3

DESIGNATION	LIMIT VALUE
Aggressive carbon dioxide (CO2)	Must not be detectable
Ammonia (NH3)	Must not be detectable
Nitrite	< 1 mg/l
Iron	< 0.3 mg/l
Manganese	< 0.05 mg/l
Sulfate	< 250 mg/l
Chloride	< 250 mg/l
COD (chemical oxygen demand)	< 40 mg/l
Microbiological growth: colony count (sulfate reducers)	< 1000/ml (must not be detectable)

Coolant additives

DESCRIPTION	RECOMMENDATION	COBES Article
Corrosion inhibitor	Multi-metal inhibitor, particularly low in salt for current-carrying induction circuits	910 0002 / 910 0003 Corrosion inhibitor KKL IK (container sizes: 100 ml/250 ml)
Biocides	Broad-spectrum biocide, free from formal-dehyde-releasing agents	910 0006 / 910 0007, Aquacid Algae-Free (container sizes: 100 ml/250 ml)
Coolant	Antifreeze with low conductivity, suitable for use down to -40°C depending on dos-age	910 0004, Cold weather additive Frost protection (container sizes: 2000 ml)

Cleaning and maintenance

The devices, as well as any external cooling systems provided, must be filled with cooling water that meets the above-mentioned water parameters before being used for the first time. We recommend adding the appropriate additives to ensure the long-term functionality of the cooling system.
Only the cooling water additives described here may be used.
These products can be obtained from COBES.

The cooling water should be checked approximately every 6 months.
The test parameters could be:

- Appearance: clear, without suspended solids.
- No odor at most a slight smell of copper inhibitor.
- pH value.
- Conductivity.
- Colony count.
- If any abnormalities are found, have a cooling water sample analyzed.

The cooling circuit must be cleaned and the water replaced once a year.

- If the limit values of the test parameters are already exceeded during the inspection cycle, the cooling circuit must be cleaned and the water replaced.

The local or national wastewater regulations at the generator installation site must be observed when disposing of the cooling water.

9.11 Dew point diagram

An air pressure of 1013 mbar was used as the basis for creating the diagram.

To prevent condensation, the cooling water temperature must be above the dew point temperature (see diagram) of the room air.

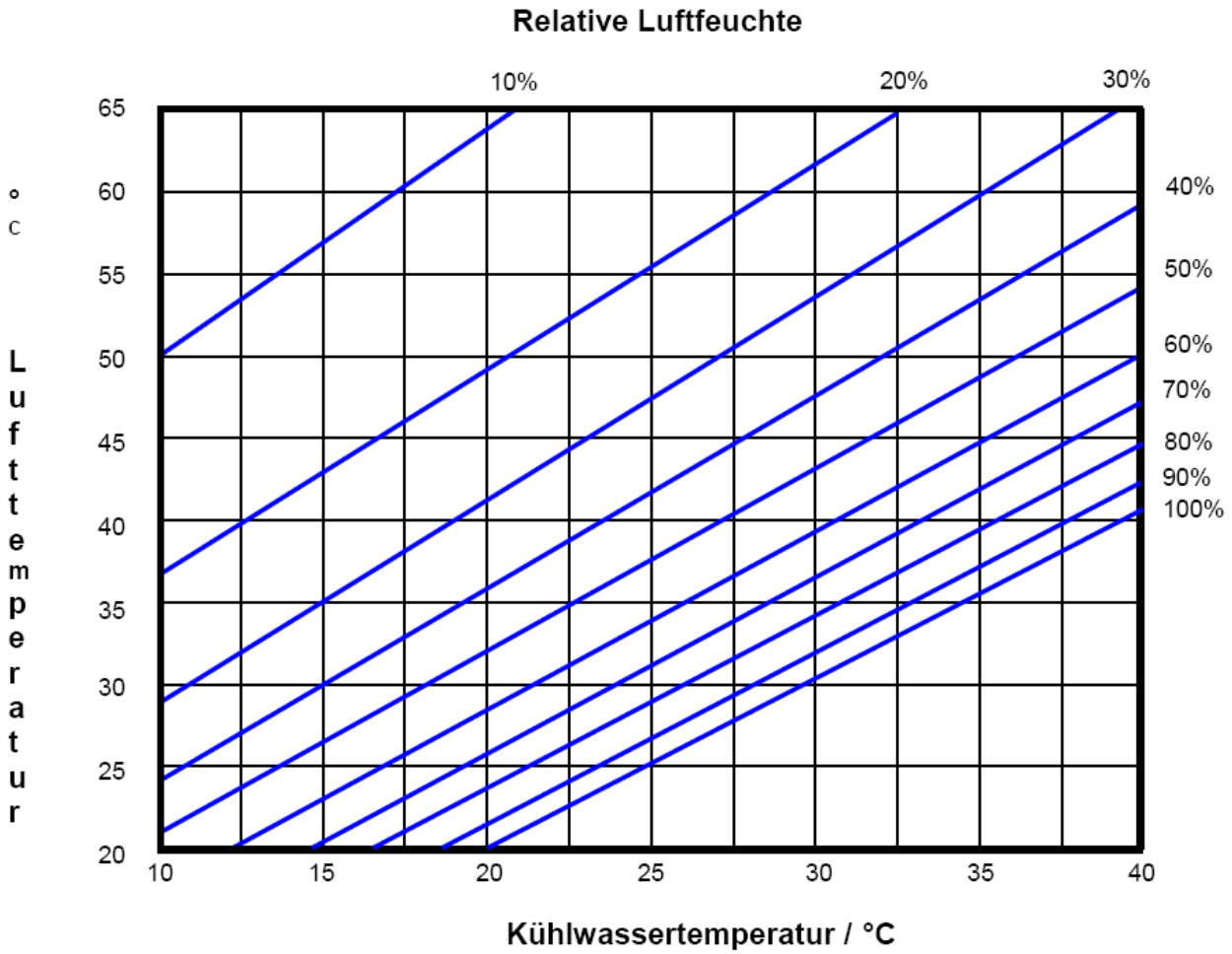
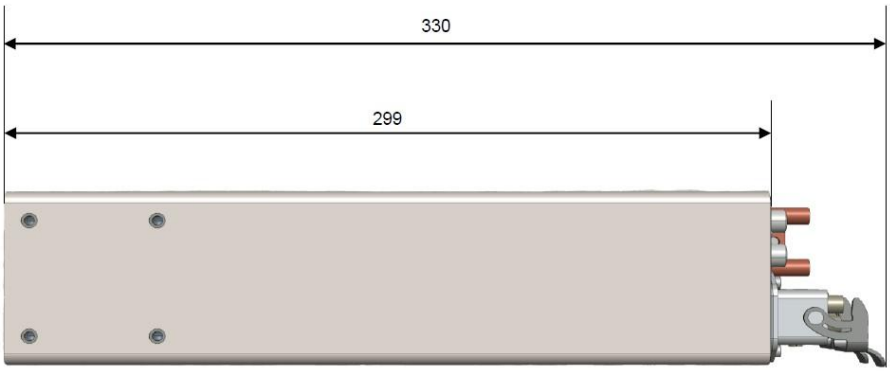
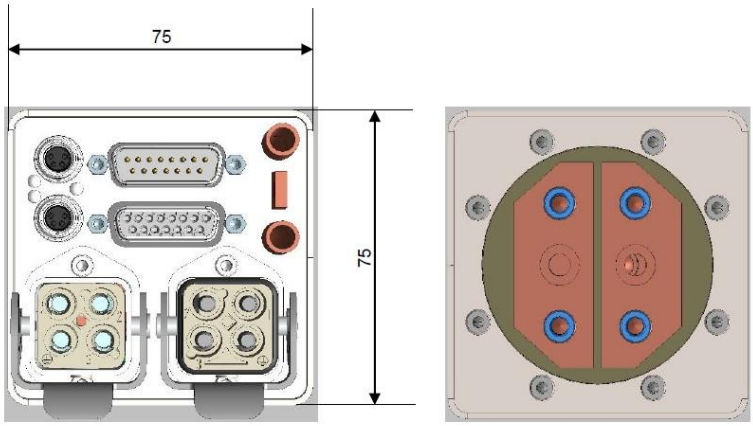


Illustration 17: Dew point diagram





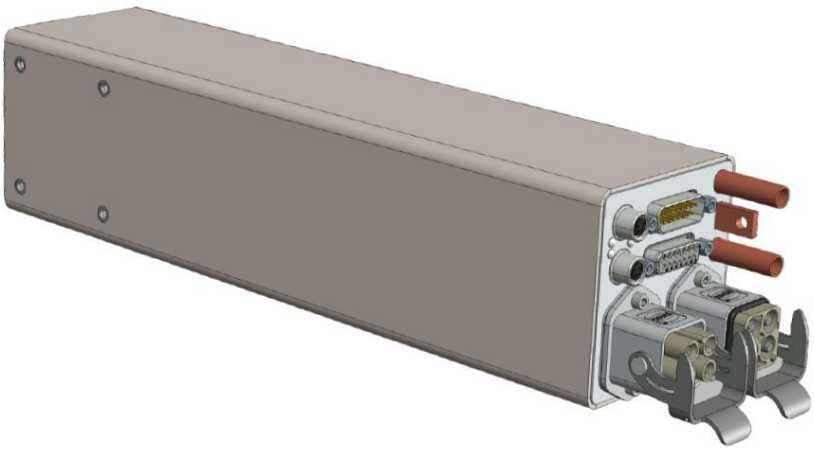


***Illustration* 18: Drawing generator**

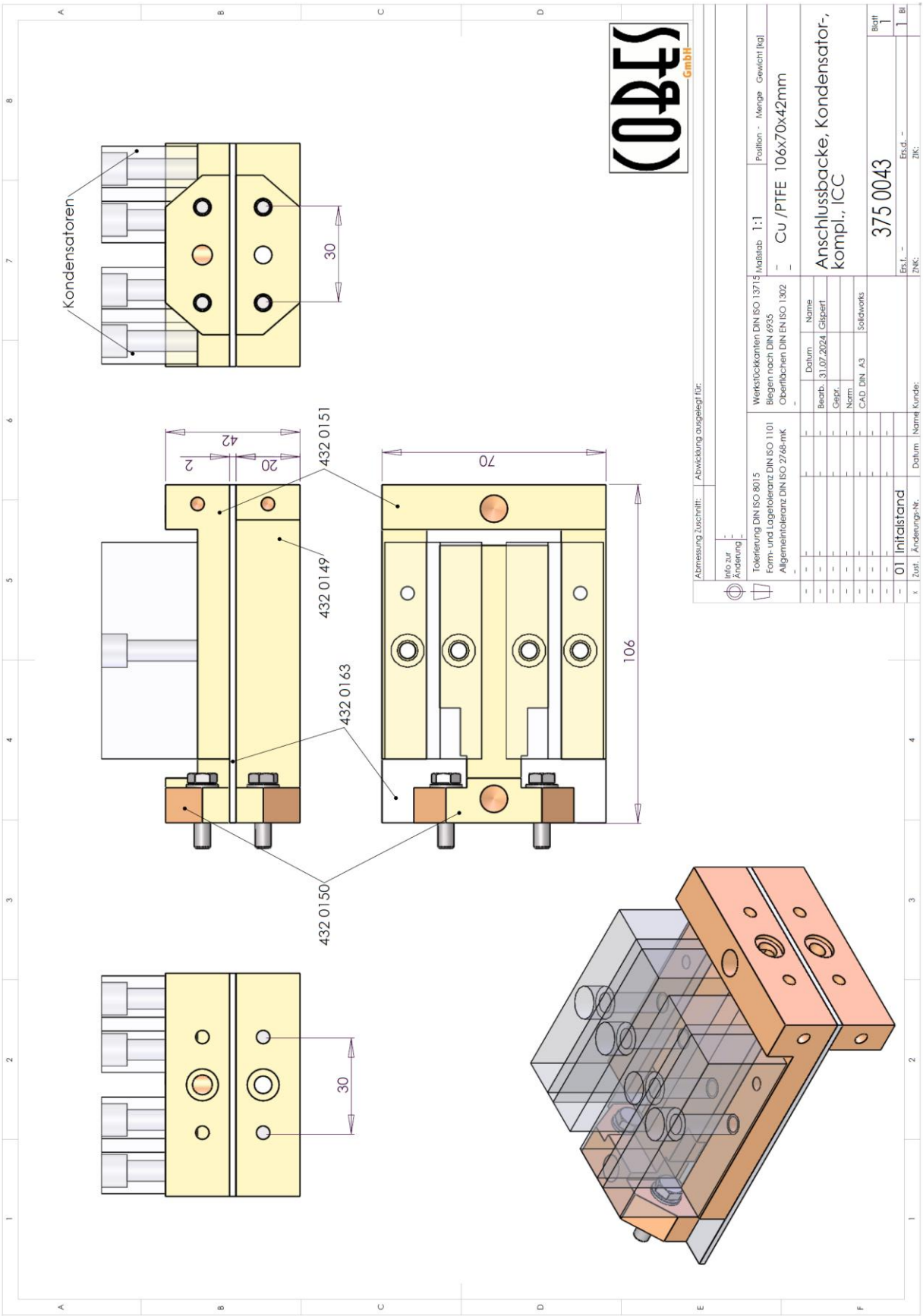


Illustration 19: Terminal block, capacitor, complete, ICC